Owrasse_Underwater_Behavior

Quick Intro

This quarto document contains the code used to visualize data from UW_Nest_Behavioral_Cleaned. The goals of this project is to understand what factors influence the mating decisions that female Ocellated wrasse make. What makes *Symphodus ocellatus* (the ocellated wrasse) interesting is their 3 different male reproductive tactics. Additionally, there are large amounts of mate choice copying from the females, so much so that a male is up significantly more likely to spawn with a female if she observes another female with him (Alonzo 2008).

The data used was collected live underwater by SCUBA divers in Corsica, Fr, in a 10 minute period.

Tidy Code

First run our library list (Install first if you don't have these packages).

```
#Install and library important packages
#install.packages('ggplot2')
#install.packages('ggeffects')
#install.packages('RColorBrewer')
#install.packages('here')
#install.packages('MASS')
#install.packages('tidyverse')
#install.packages('janitor')
#install.packages('readxl')
#install.packages('brms')
library(ggeffects)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```r
library(here)
```

```
here() starts at /Users/jacobchung/coding_projects/Owrasse_Underwater_Behavioral_Sheet/Explor
```

```r
library(ggplot2)
library(readxl)
library(dplyr)
library(brms)
```

```
Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').

Attaching package: 'brms'

The following object is masked from 'package:stats':

    ar
```

```r
library(RColorBrewer)
library(tibble)
library(MASS)
```

```
Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

Importing the data sheet

```r
behavior_sheet <- read_excel("UW_Nest_Behavioral_Sheets_Cleaned.xlsx")
head(behavior_sheet)
```

```
# A tibble: 6 x 70
  Date                `Nest Name` `Start Time`        `Obs Length` `NM Mark`
  <dttm>              <chr>       <dttm>              <chr>        <chr>
1 2025-05-21 00:00:00 4           1899-12-31 10:19:00 10           UM
2 2025-05-22 00:00:00 OSL SS      1899-12-31 11:00:00 10           UM
3 2025-05-23 00:00:00 8           1899-12-31 10:14:00 10           UM
4 2025-05-25 00:00:00 25          1899-12-31 10:16:00 10           UM
5 2025-05-25 00:00:00 FOSR ZZ     1899-12-31 10:54:00 10           UM
6 2025-05-26 00:00:00 OSR YY      1899-12-31 10:12:00 10           UM
# i 65 more variables: `SAT Mark` <chr>, Observer <chr>,
#   `Observation type` <chr>, `Average SNK` <dbl>, `# NM in` <dbl>,
#   `# NM near` <dbl>, `# NM far` <dbl>, `# SAT Present` <dbl>,
#   `# Sat Absent` <dbl>, `Female visits by NM` <dbl>,
#   `Female visits by SAT passed  to NM` <dbl>, `Female visit by SAT` <dbl>,
#   `Female visit by SNK` <dbl>, `Female spawning with NM brought by NM` <dbl>,
#   `Female spawning with only SAT brought by NM` <dbl>, ...
```

Separating the date column and placing it near the original Date column for filtering ease later on.

```r
behavior_sheet_date <- as.Date(behavior_sheet$Date)
behavior_sheet$Date <- behavior_sheet_date
behavior_sheet_sep <- behavior_sheet %>%
  mutate(Year = year(Date)) %>%
  mutate(Month = month(Date)) %>%
  mutate(Day = day(Date)) %>%
  relocate(Year, .after = Date) %>%
  relocate(Month, .after = Year) %>%
  relocate(Day, .after = Month)
head(behavior_sheet_sep)
```

```
# A tibble: 6 x 73
  Date        Year Month   Day `Nest Name` `Start Time`        `Obs Length`
  <date>     <dbl> <dbl> <int> <chr>       <dttm>              <chr>
1 2025-05-21  2025     5    21 4           1899-12-31 10:19:00 10
2 2025-05-22  2025     5    22 OSL SS      1899-12-31 11:00:00 10
3 2025-05-23  2025     5    23 8           1899-12-31 10:14:00 10
4 2025-05-25  2025     5    25 25          1899-12-31 10:16:00 10
5 2025-05-25  2025     5    25 FOSR ZZ     1899-12-31 10:54:00 10
6 2025-05-26  2025     5    26 OSR YY      1899-12-31 10:12:00 10
# i 66 more variables: `NM Mark` <chr>, `SAT Mark` <chr>, Observer <chr>,
#   `Observation type` <chr>, `Average SNK` <dbl>, `# NM in` <dbl>,
#   `# NM near` <dbl>, `# NM far` <dbl>, `# SAT Present` <dbl>,
#   `# Sat Absent` <dbl>, `Female visits by NM` <dbl>,
#   `Female visits by SAT passed  to NM` <dbl>, `Female visit by SAT` <dbl>,
#   `Female visit by SNK` <dbl>, `Female spawning with NM brought by NM` <dbl>,
#   `Female spawning with only SAT brought by NM` <dbl>, ...
```

```r
#Fix SAT submission to NM from chr to num
sat_sub <- as.numeric(behavior_sheet_sep$`SAT submission to NM`) #Creates a vector with the s
```

```
Warning: NAs introduced by coercion
```

```r
behavior_sheet_sep$`SAT submission to NM` <- sat_sub #put it back into the tibble
#Do the same for SNK Submission to SAT
snk_sub <- as.numeric(behavior_sheet_sep$`SNK submission to SAT`)
```

```
Warning: NAs introduced by coercion
```

```r
behavior_sheet_sep$`SNK submission to SAT` <- snk_sub
#Same for NM aggression to SAT
NM_aggsat <- as.numeric(behavior_sheet_sep$`NM aggression to SAT`)
```

```
Warning: NAs introduced by coercion
```

```r
behavior_sheet_sep$`NM aggression to SAT` <- NM_aggsat
```

Create a tibble with cleaned names

This chunk will be useful when creating and running models later on because it "standardizes" the column names.

```r
behavior_sheet_cln <- behavior_sheet_sep |>
  #Uses the Janitor package to standardize the names
  clean_names()

#Turn the year variable into a character for ease of comparison later on
year_date <- as.character(behavior_sheet_cln$year) #Creates a vector with years as chr
behavior_sheet_cln$year <- year_date #creates a new variable for ease of graphing later on
```

Year totals

For ease of comparison between years and months, this code will add up the total number of nests for each year.

```r
yearly_totals <- behavior_sheet_sep |>
  group_by(Year) |>
  summarise(
    # count the number of nests in each year
    total_nests = n(),
    # take the avg of the "Average SNK" column for each year and make new column
    avg_SNKR = mean(`Average SNK`, na.rm = TRUE),
          # include only numeric columns
    across(where(is.numeric) &
             # exclude Month, Day, and Average SNKR column
             !c(Month, Day, `Average SNK`),
           # add up remaining totals per year for each column
           ~ sum(.x, na.rm = TRUE)))
```

```r
yearly_totals
```

```
# A tibble: 7 x 64
   Year total_nests avg_SNKR `# NM in` `# NM near` `# NM far` `# SAT Present`
  <dbl>       <int>    <dbl>     <dbl>       <dbl>      <dbl>           <dbl>
1  2017           8     3.65        94          60          6              79
2  2018          22     3.15       230         192         18             204
3  2019          38     2.60       342         397         21             317
4  2022           9     3.53       100          79          1              86
5  2023          21     3.85       258         159          3             210
6  2024          19     2.88       220         152          8             189
7  2025          24     3.60       347         130          3             240
# i 57 more variables: `# Sat Absent` <dbl>, `Female visits by NM` <dbl>,
#   `Female visits by SAT passed  to NM` <dbl>, `Female visit by SAT` <dbl>,
#   `Female visit by SNK` <dbl>, `Female spawning with NM brought by NM` <dbl>,
#   `Female spawning with only SAT brought by NM` <dbl>,
#   `Female spawning Sat passed to NM` <dbl>,
#   `Female spawning Sat passed to NM but only SAT spawn` <dbl>,
#   `Female spawning Sat brought` <dbl>, ...
```

```r
#Turn the year variable into a character for ease of comparison later on
year_date <- as.character(yearly_totals$Year) #Creates a vector with years as chr
yearly_totals$Year_axis <- year_date #creates a new variable for ease of graphing later on

head(yearly_totals) #Double Check
```

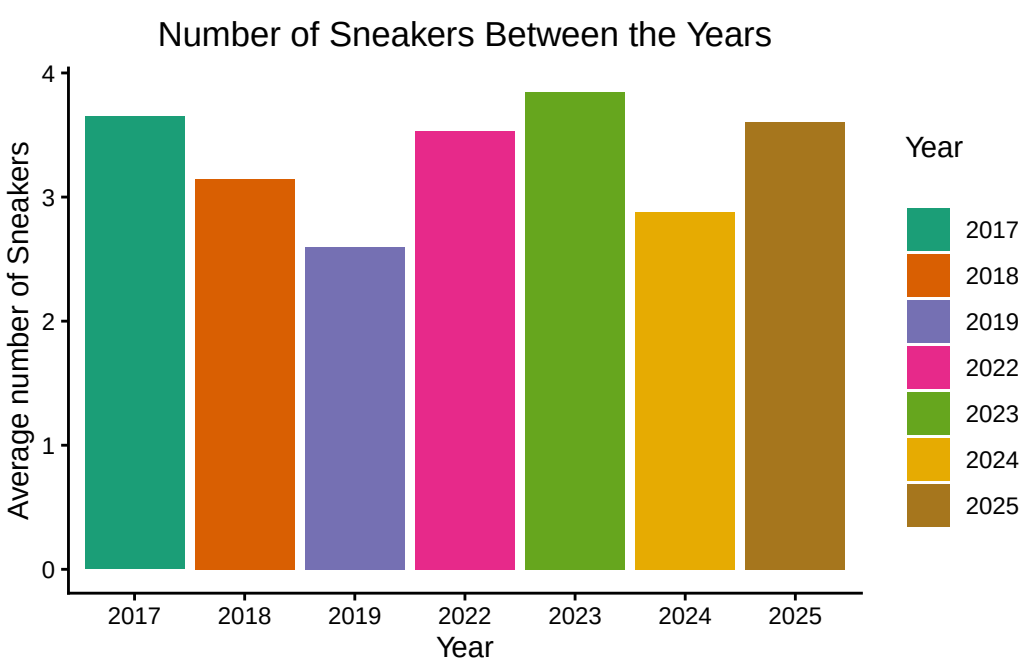
```
# A tibble: 6 x 65
   Year total_nests avg_SNKR `# NM in` `# NM near` `# NM far` `# SAT Present`
  <dbl>       <int>    <dbl>     <dbl>       <dbl>      <dbl>           <dbl>
1  2017           8     3.65        94          60          6              79
2  2018          22     3.15       230         192         18             204
3  2019          38     2.60       342         397         21             317
4  2022           9     3.53       100          79          1              86
5  2023          21     3.85       258         159          3             210
6  2024          19     2.88       220         152          8             189
# i 58 more variables: `# Sat Absent` <dbl>, `Female visits by NM` <dbl>,
#   `Female visits by SAT passed  to NM` <dbl>, `Female visit by SAT` <dbl>,
#   `Female visit by SNK` <dbl>, `Female spawning with NM brought by NM` <dbl>,
#   `Female spawning with only SAT brought by NM` <dbl>,
#   `Female spawning Sat passed to NM` <dbl>,
```

```
#   `Female spawning Sat passed to NM but only SAT spawn` <dbl>,
#   `Female spawning Sat brought` <dbl>, ...
```

Exploratory graphs looking between the years

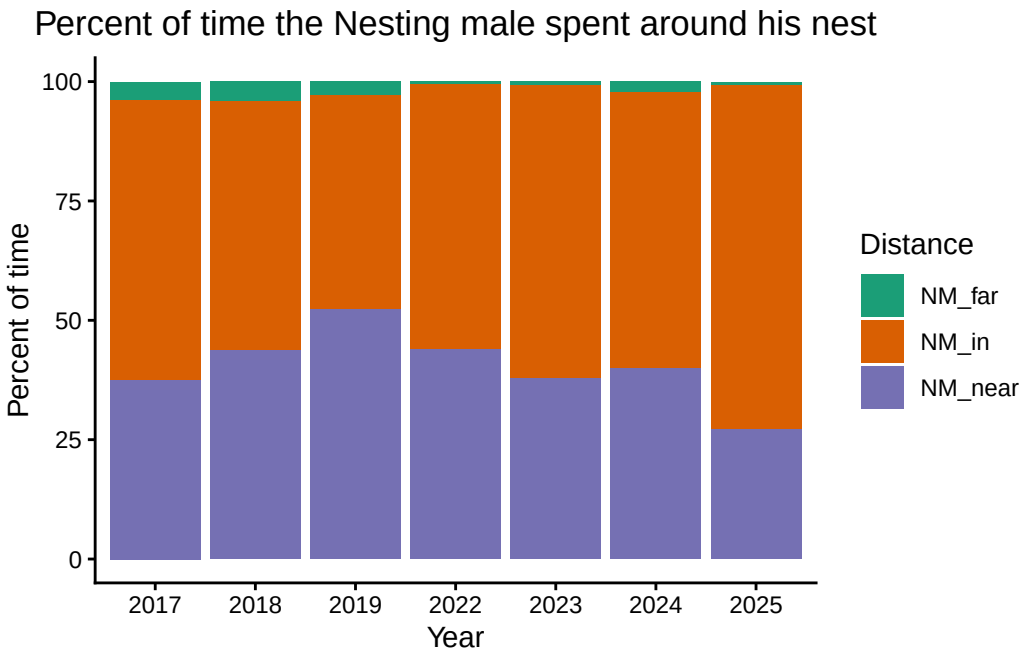
Average sneakers between years

```
avg_snk_years <- yearly_totals |>
  ggplot(aes(x = Year_axis, y = avg_SNKR, fill = Year_axis)) + #gives ggplot the dimensions
  geom_bar(stat = 'identity') + #makes the plot a bar/column chart
  labs(x = 'Year',
       y = 'Average number of Sneakers',
       title = "Number of Sneakers Between the Years",
       fill = 'Year
       ') + #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars
avg_snk_years
```



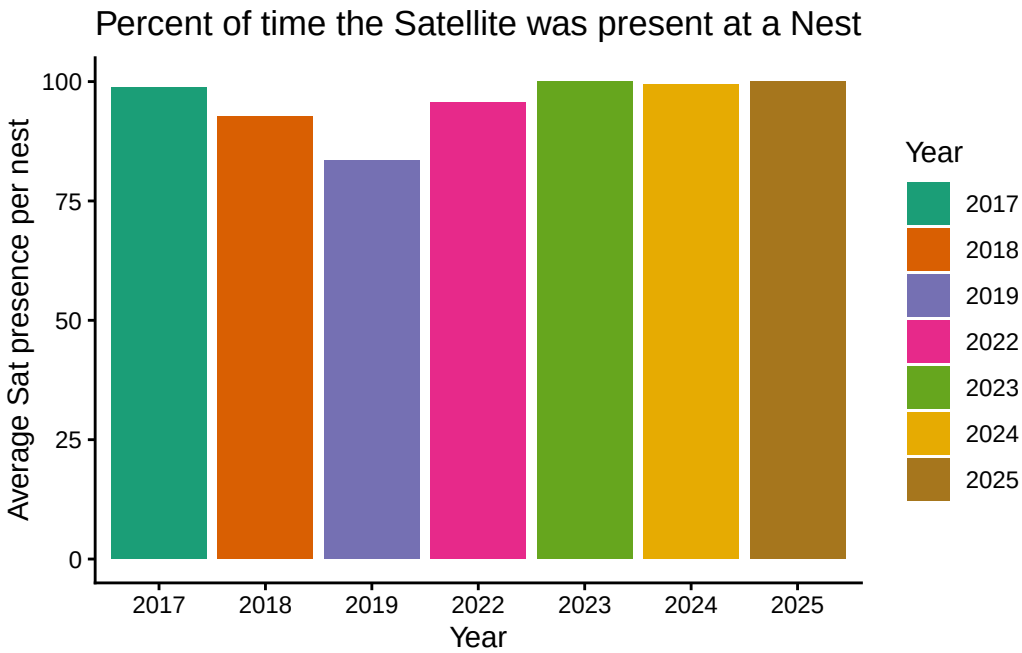
Average Nesting Male Time in/near nest between years

```
NM_in_near <- yearly_totals |>
  group_by(Year_axis) |> #Focusing on per year
  summarise(
    #Get the total amount of time that the NM was recorded
    total_time = round(sum(`# NM in` + `# NM near` + `# NM far`), digits = 4),
    #Get the percent of time the NM was in his nest.
    NM_in = round((`# NM in`/total_time)*100, digits = 4),
    #Get the percent of time the NM was near his nest.
    NM_near = round((`# NM near`/total_time)*100, digits = 4),
    #Get the percent of time the NM was far from his nest
    NM_far = round((`# NM far`/total_time)*100, digits = 4))

NM_in_near.long <- NM_in_near %>%
  pivot_longer(cols = -c(Year_axis, total_time), #Take all tibble coloumns except Years and t
               names_to = "distance", # Put the percent coloumns into their own
               values_to = "perc_time") #Put previous' columns into a new one


NM_in_near.long %>%
  ggplot(aes(Year_axis, perc_time, fill = distance)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
    labs(x = 'Year',
       y = 'Percent of time',
       title = "Percent of time the Nesting male spent around his nest",
       fill = 'Distance') + #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars
```



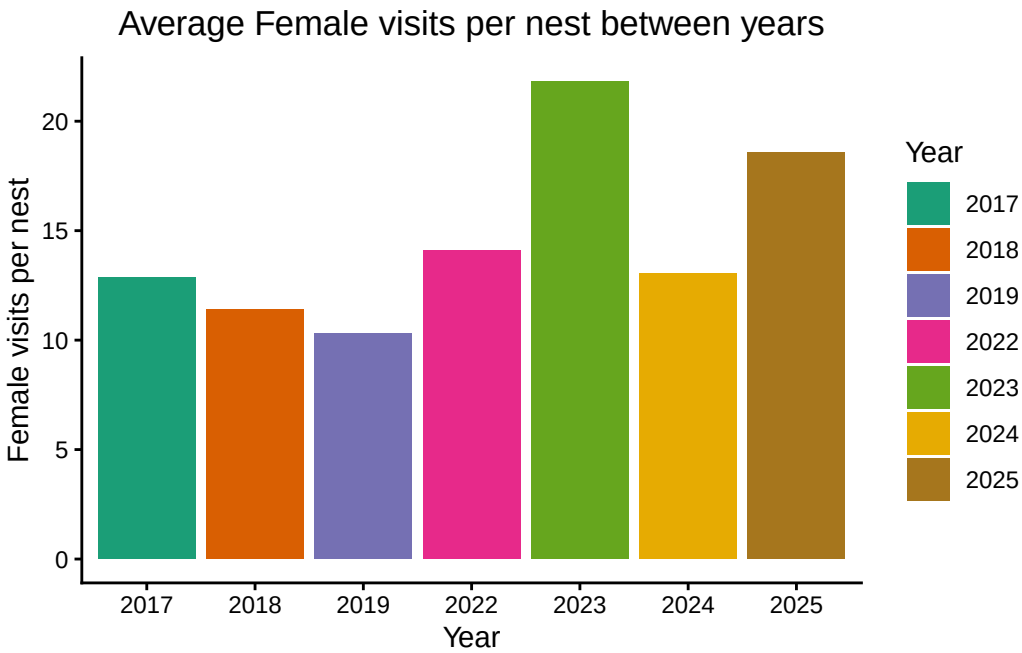
Average SAT Presence at the Nest between years per nest

```
SAT.presence <- yearly_totals |>
  mutate('SatP_perN' = (`# SAT Present`/total_nests)*10, .after = 7) #creates the column wit

SAT.presence.plot <- SAT.presence |>
  ggplot(aes(Year_axis, SatP_perN, fill = Year_axis)) + #gives ggplot dimensions
  geom_bar(stat = 'identity') + #makes the plot a bar/column chart
  labs(x = 'Year',
       y = 'Average Sat presence per nest',
       title = "Percent of time the Satellite was present at a Nest",
       fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars

SAT.presence.plot
```



Average Female Visits per nest between years

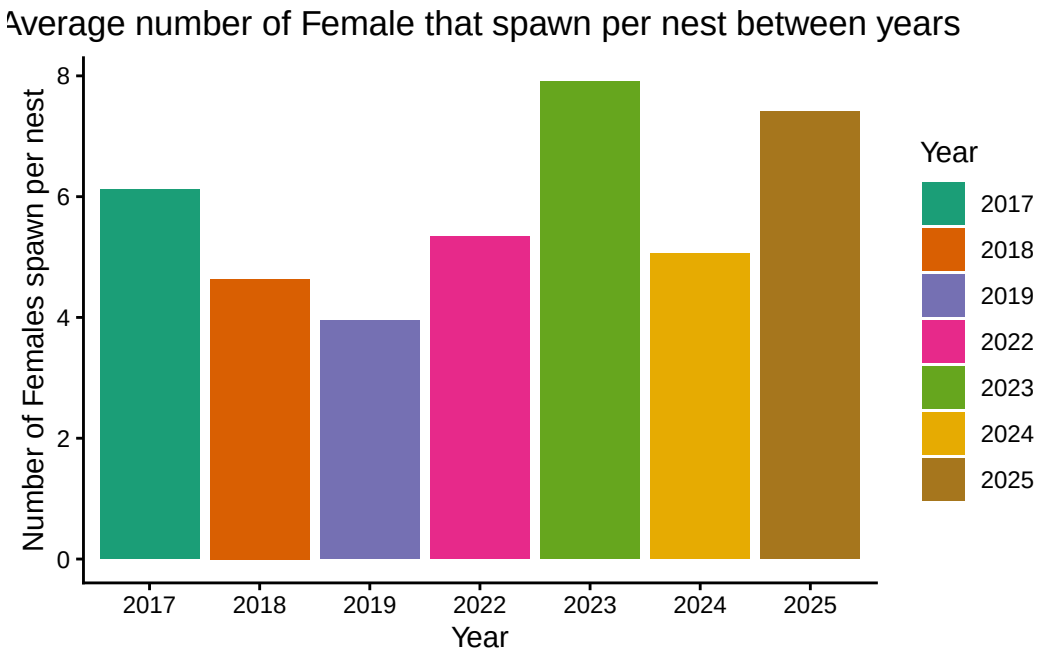
```
T_female_visits <- yearly_totals |>
  #creates a variable that is the average female visits per nest
  mutate('visits_p_nest' = (`Total Female Visits`)/total_nests) |>
  ggplot(aes(Year_axis, visits_p_nest, fill = Year_axis)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
  labs(x = 'Year',
      y = 'Female visits per nest',
      title = "Average Female visits per nest between years",
      fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars


T_female_visits
```



Average Female spawns between years

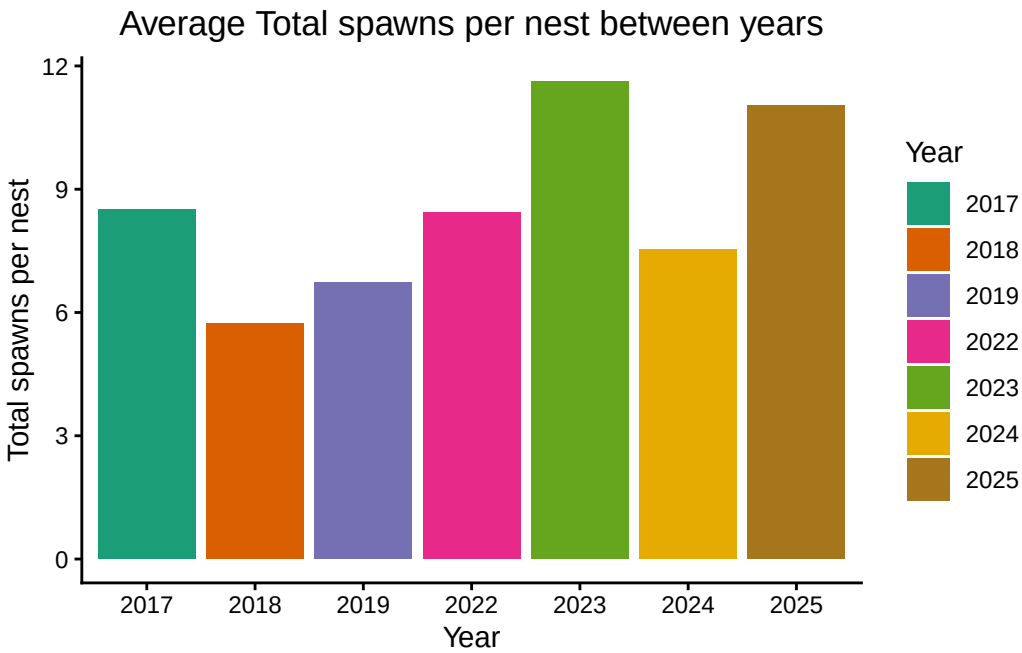
This is different from the next graph, because this is the number of females that spawned. So each female is only counted once, while females can spawn multiple times (which went to the total variable).

```
A_female_spawns <- yearly_totals |>
  #Gets average total female spawns between the years
  mutate('avg_fspawns_nest' = (`Total Female Spawning`)/total_nests) |>
  ggplot(aes(Year_axis, avg_fspawns_nest, fill = Year_axis)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
  labs(x = 'Year',
      y = 'Number of Females spawn per nest',
      title = "Average number of Female that spawn per nest between years",
      fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars

A_female_spawns
```



Average Total spawns between years

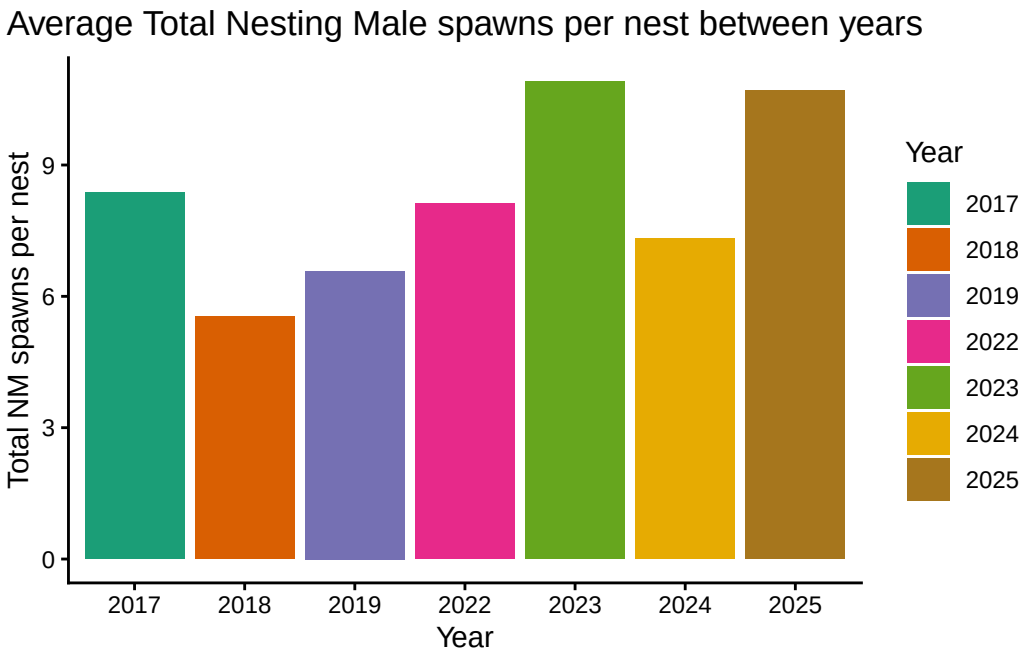
```
T_total_spawns <- yearly_totals |>
  #Creates the column of total spawns per nest
  mutate('total_spawns_p_nest' = (`Total Spawns`)/total_nests) |>
  #Creates the plot
  ggplot(aes(Year_axis, total_spawns_p_nest, fill = Year_axis)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
  labs(x = 'Year',
      y = 'Total spawns per nest',
      title = "Average Total spawns per nest between years",
      fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars


T_total_spawns
```



Average Total Nesting Male Spawns between years

This graph adds up all the columns that indicate Nesting male spawns and compares the averages between the years

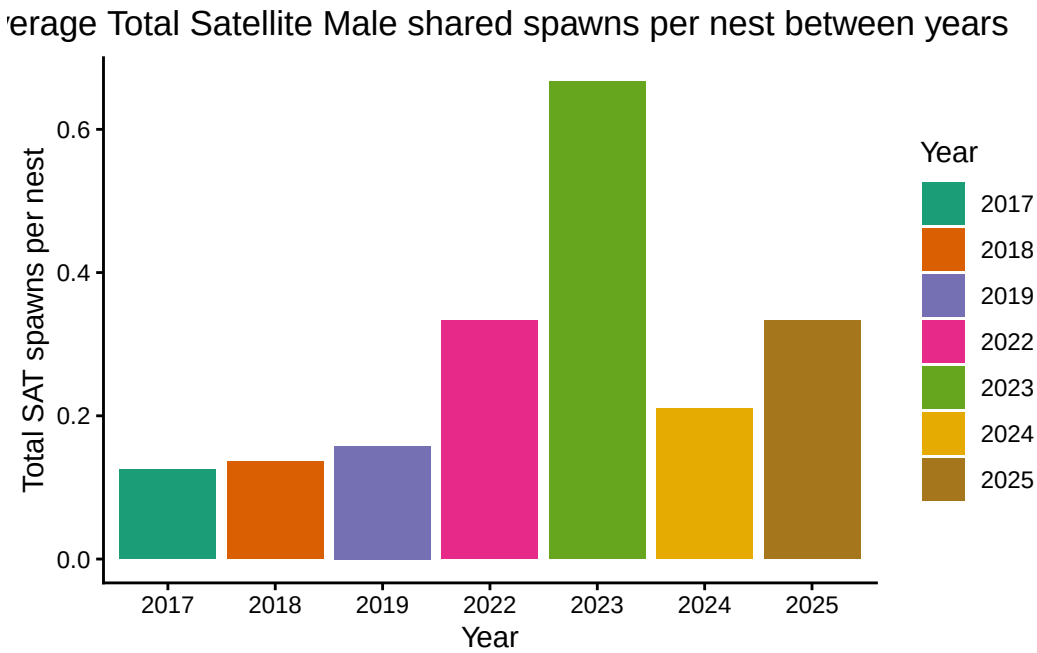
```
T_NM_spawns <- yearly_totals |>
  #Average total nesting male spawns
  mutate('total_NM_spawns' = (`Total Spawns NM brought` + `Total spawns SAT passed to NM` +
  ggplot(aes(Year_axis, total_NM_spawns, fill = Year_axis)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
  labs(x = 'Year',
      y = 'Total NM spawns per nest',
      title = "Average Total Nesting Male spawns per nest between years",
      fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars

T_NM_spawns
```



Total Satellite Spawns between years

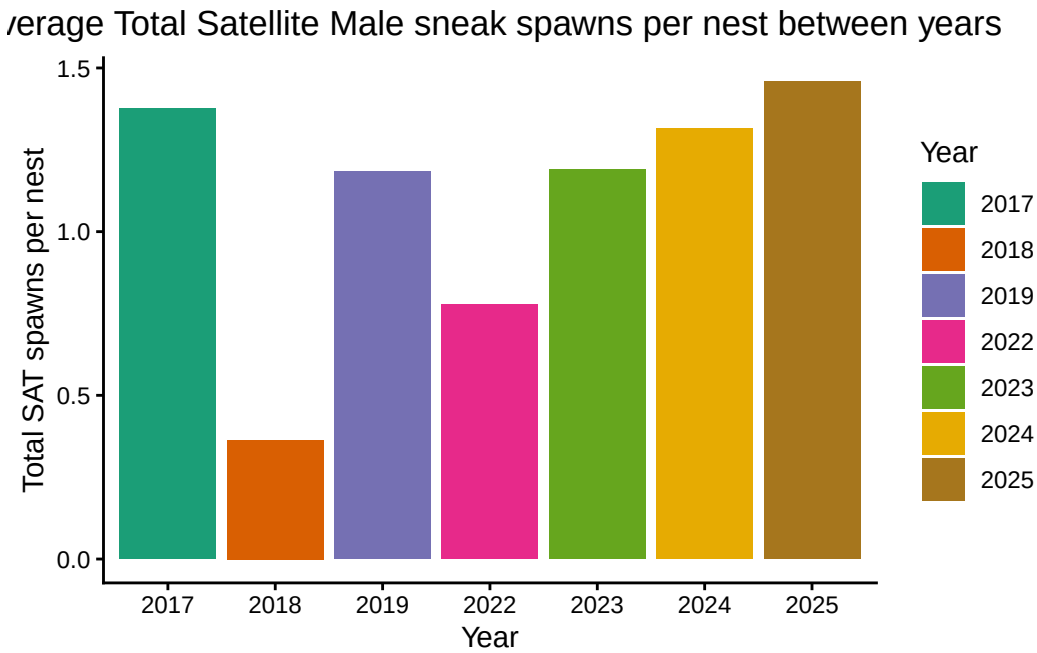
This graph adds up all the columns that indicate Satellite spawns and compares between the years. Note, this is counting when the Satellite is *allowed* to spawn at the nest by the Nesting male.

```
T_SAT_spawns <- yearly_totals |>
  #Creates a variable that is the total amount of SAT spawns
  mutate('T_SAT_spawn' = (`Total spawns NM brought, SAT only spawn` + `Total spawns SAT passe
  ggplot(aes(Year_axis, T_SAT_spawn, fill = Year_axis)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
  labs(x = 'Year',
      y = 'Total SAT spawns per nest',
      title = "Average Total Satellite Male shared spawns per nest between years",
      fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars

T_SAT_spawns
```



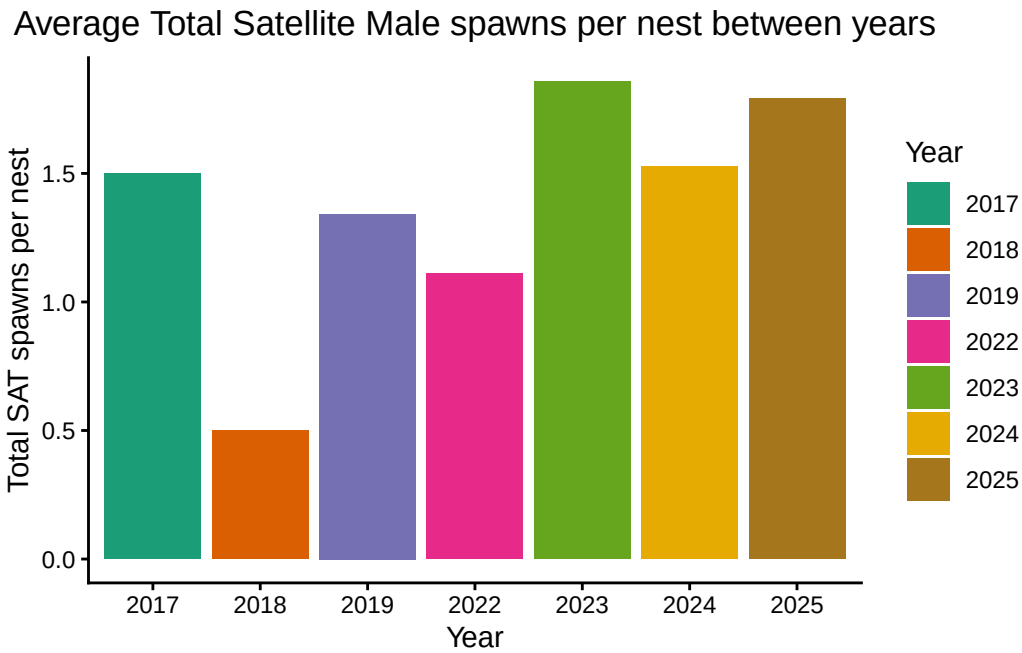
This graph will count Satellite *sneak* spawns (note this does include sneaker sneaks).

```
T_SAT_sneaks <- yearly_totals |>
  #Creates a variable that is all the sneak spawns of the SAT
  mutate('T_SAT_sneak' = (`Female visit NM brought, Sneaks SAT + SNK` + `Female visit NM brou
  ggplot(aes(Year_axis, T_SAT_sneak, fill = Year_axis)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
  labs(x = 'Year',
      y = 'Total SAT spawns per nest',
      title = "Average Total Satellite Male sneak spawns per nest between years",
      fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars

T_SAT_sneaks
```



This graph will do total SAT spawns, both shared and sneak

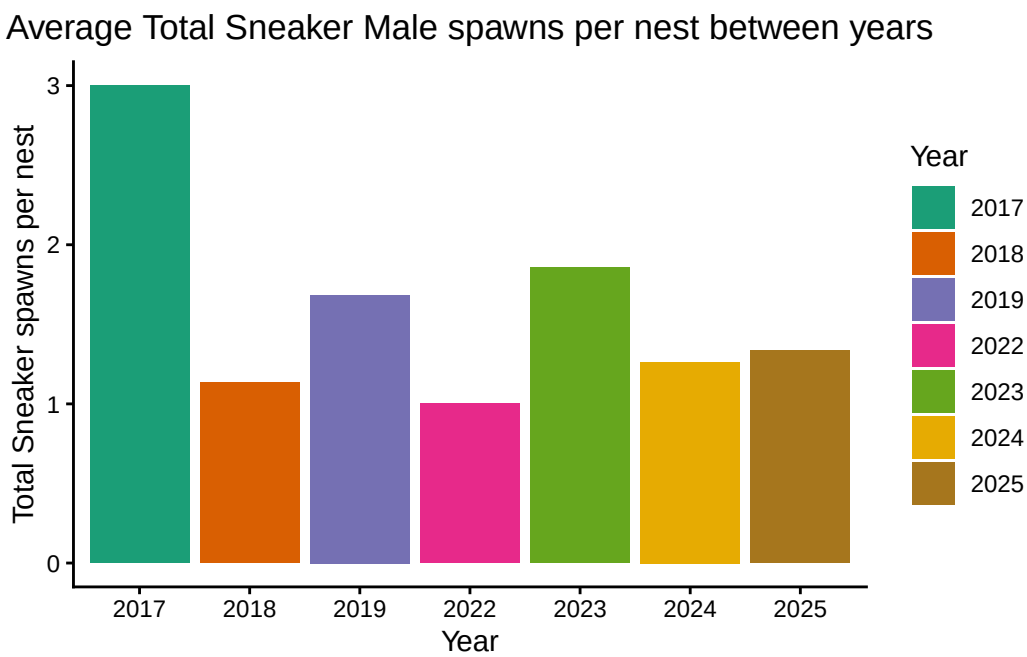
```
T_SAT_spawns_both <- yearly_totals |>
  #Creates a variable that is the total of both types of SAT spawns
  mutate('T_SAT_spawn_both' = (`Total spawns NM brought, SAT only spawn` + `Total spawns SAT
  ggplot(aes(Year_axis, T_SAT_spawn_both, fill = Year_axis)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
  labs(x = 'Year',
      y = 'Total SAT spawns per nest',
      title = "Average Total Satellite Male spawns per nest between years",
      fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars

T_SAT_spawns_both
```



Total Sneaker Spawns between years

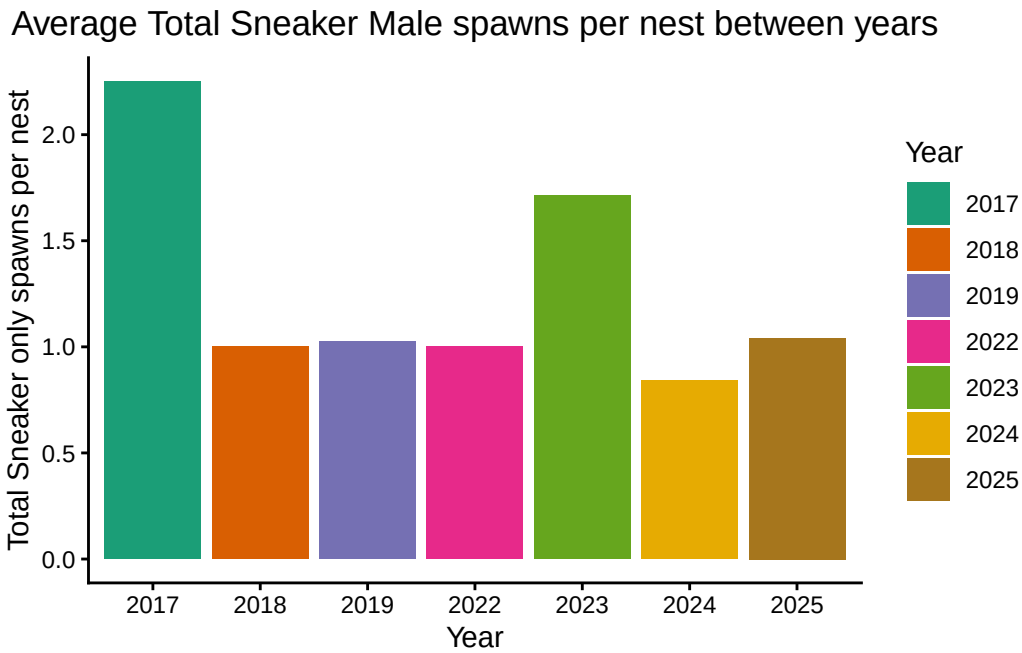
This graph adds up all the columns that indicate Sneaker male spawns and compares between the years. Note, some sneaks are including SAT sneaks

```
T_SNK_spawns <- yearly_totals |>
  #Average Total sneaker spawns per nest, includes SAT variables
  mutate('total_sneaks' = (`Total Spawns SNK brought, SNK only spawn` + `Female visit NM brou
  ggplot(aes(Year_axis, total_sneaks, fill = Year_axis)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
  labs(x = 'Year',
      y = 'Total Sneaker spawns per nest',
      title = "Average Total Sneaker Male spawns per nest between years",
      fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars

T_SNK_spawns
```



Graph is for sneakers only

```
T_SNK_spawns_only <- yearly_totals |>
  #Average total snaker spawns per nest, sneaker
  mutate('total_sneaks_only' = (`Total Spawns SNK brought, SNK only spawn`+ `Female visit NM
  ggplot(aes(Year_axis, total_sneaks_only, fill = Year_axis)) + #gives ggplot dimensions
  geom_col() + #makes the plot a column chart
  labs(x = 'Year',
      y = 'Total Sneaker only spawns per nest',
      title = "Average Total Sneaker Male spawns per nest between years",
      fill = 'Year') +  #edits axis and title labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) + #Moves title to the middle
  scale_fill_brewer(palette = "Dark2") #Colors the bars

T_SNK_spawns_only
```



Other, Not Year Specific, Exploratory graphs and stats

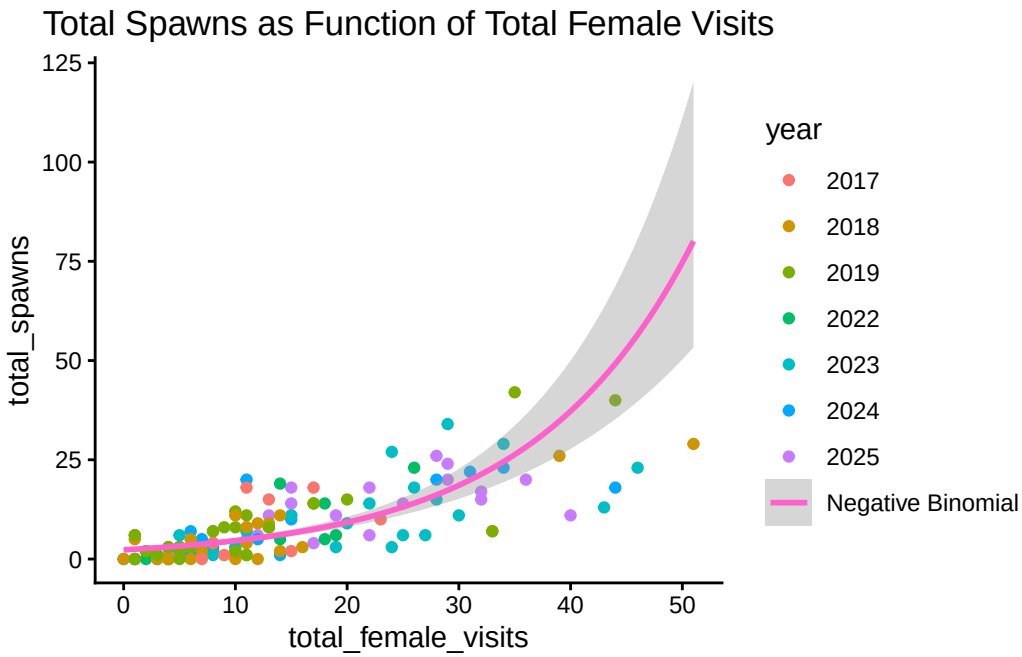
Female visits and Female Spawns

Presumably, more visits should = more spawns is this true? We use total female visits instead of total female spawning because female visits counts the females that leave the nest without spawning, while total female spawning only counts individuals who spawned.

This is the exploratory graph

```
f_visits_spawns <- behavior_sheet_cln |>
  ggplot(aes(total_female_visits, total_spawns, color = year )) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth(aes(color = 'Negative Binomial'), se = TRUE, method = 'glm.nb') +  #Add a linea
  labs(title = 'Total Spawns as Function of Total Female Visits') +
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle


f_visits_spawns
```

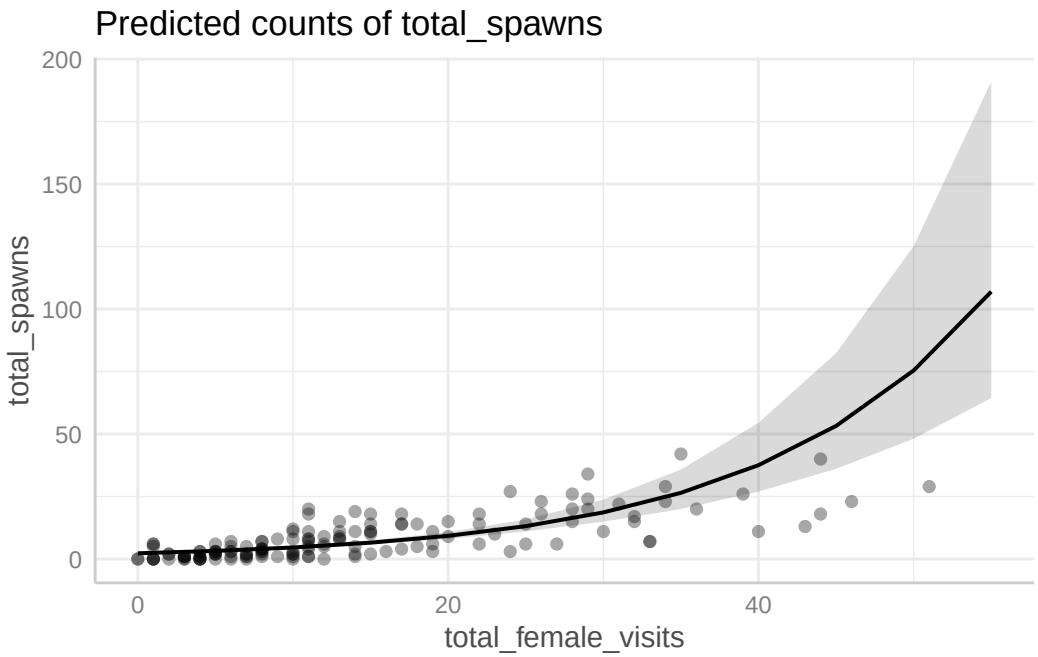
```
`geom_smooth()` using formula = 'y ~ x'
```



This is the exploratory linear regression model

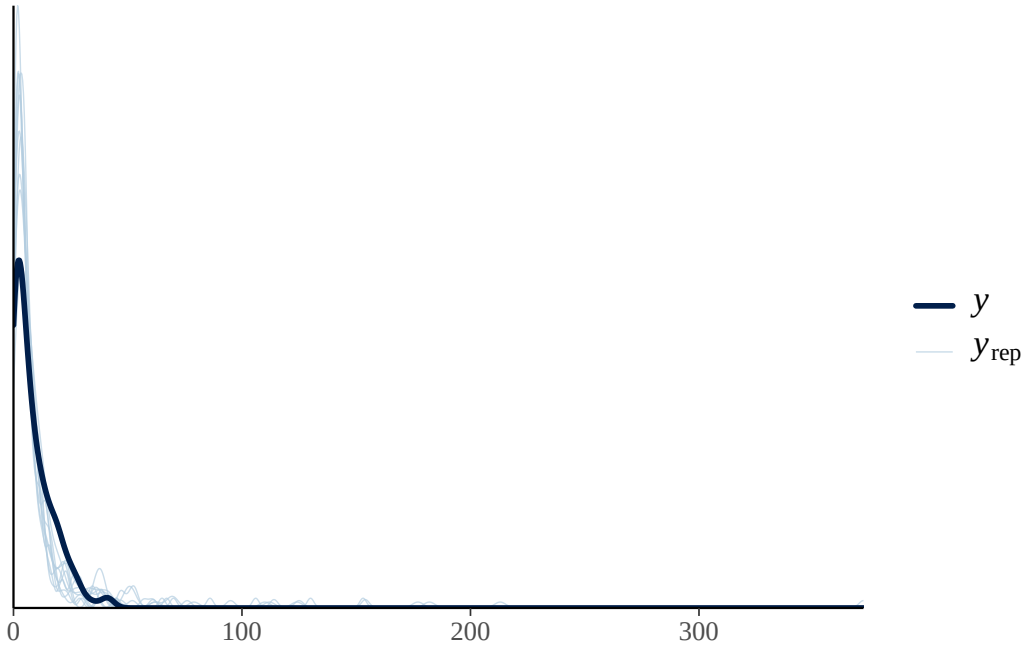
```
#Total spawns as a function of total female visits
f.tspawns.tvisits.nbmdl <-
      brm(data = behavior_sheet_cln, #Gives the model the tibble
      family = negbinomial(), #Tells the model what distribution to use
      total_spawns ~ total_female_visits, #specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Specifying Markov chain parameters
      seed = 4, #Determines which random numbers will get sampled
      file = 'Exploratory model outputs/f.tspawns.tvisits.nbmdl') #Saves the model as an obje

preds <- predict_response(f.tspawns.tvisits.nbmdl)
plot(preds, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```




```
pp_check(f.tspawns.tvisits.nbmdl)
```

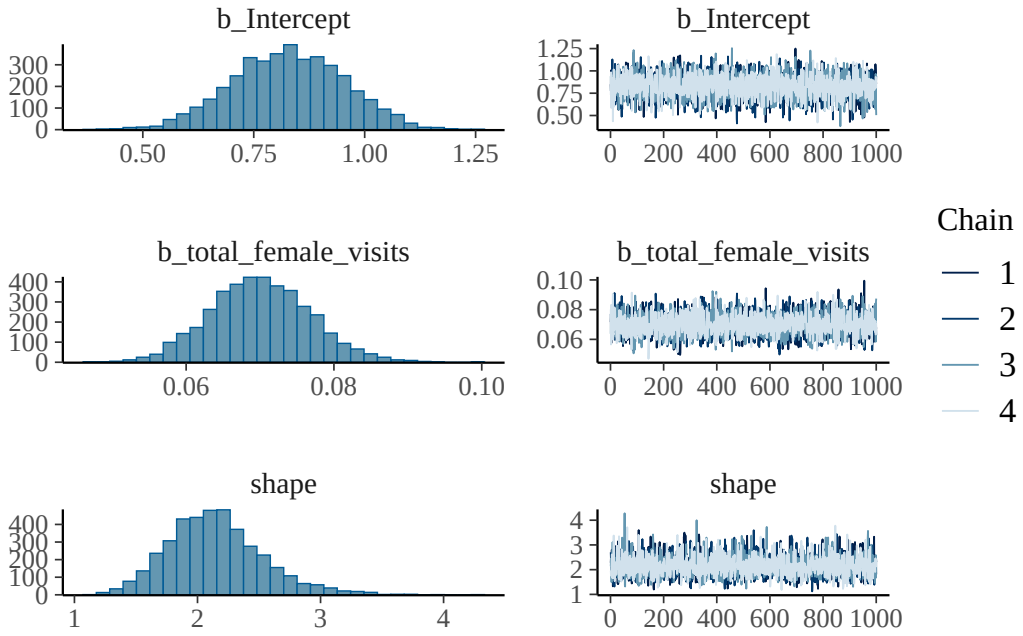
```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```



```r
summary(f.tspawns.tvisits.nbmdl) #Gives us model summary
```

```
 Family: negbinomial
  Links: mu = log
Formula: total_spawns ~ total_female_visits
   Data: behavior_sheet_cln (Number of observations: 141)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                    Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept               0.83      0.13     0.58     1.07 1.00     3471     3257
total_female_visits     0.07      0.01     0.06     0.08 1.00     4151     2707

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
shape     2.15      0.39     1.48     3.04 1.00     3012     2809

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(f.tspawns.tvisits.nbmdl) #plots the Markov chains
```



Given our model, we can suggest that for every female visit, there is a 0.07 increase in the total spawns at that nest. Our 95% credibility interval gives us 0.06-0.08, allowing us to suggest that this effect is different from zero. This matches our expectation because with every visit to a nest, we would expect a general increase in the total number of spawns at a nest (however not complete because remember, some females don't spawn).

Note:

At lower numbers of female visits, there seem to be much less variation than there is when there are high number of female visits. This is both intuitive, and counter intuitive. Intuitive in the sense that, at lower nest visits, there are less number of spawns because there are lower number of females visiting; while at high nest visits, there is more opportunity for females to not spawn and leave. However, it is counter intuitive because, previous research suggests that mate choice copying is the primary driver of female spawning decisions, so we would expect that higher visitation rates (meaning more females around the nest in general, meaning higher chances of females observing other females spawning) we would have spawning rates that are *consistent*.

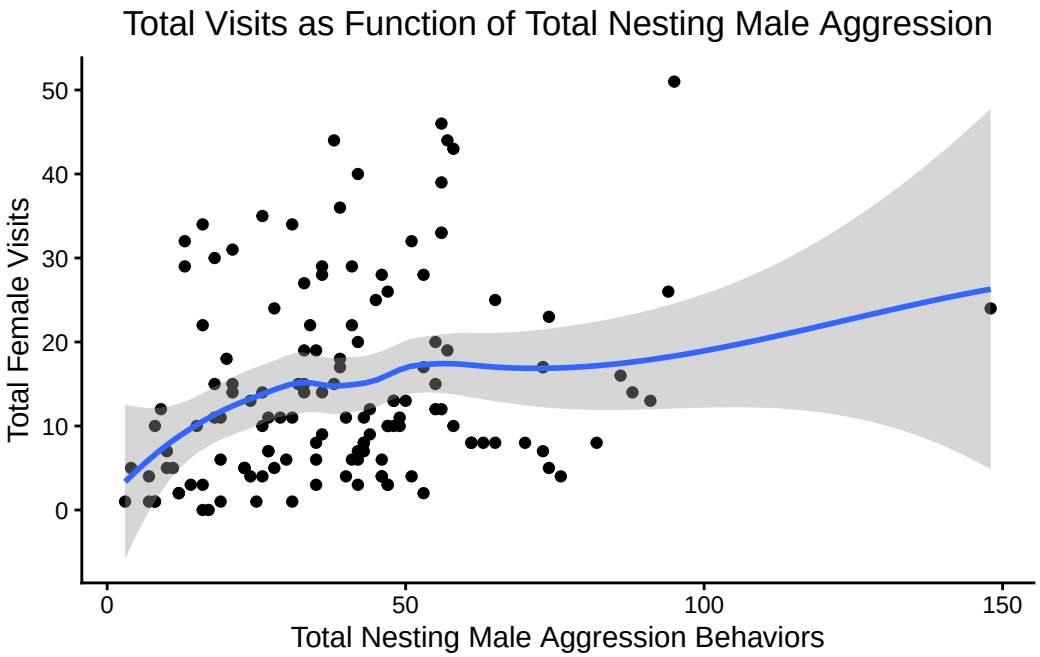
Nesting Male aggression effect on Female visits and spawns

This adds up all the columns related to Nesting male aggression (Except aggression towards/away from nest, this will be graphed later) and looks for any correlation with female visits and spawns.

Female visits as a function of Nesting Male aggression

```
fvisits.nmagg <- behavior_sheet_cln |>
  #creates a variable that is a total of all NM agression
  mutate('total_nm_agg' = nm_aggression_to_sat + nm_aggression_to_snk + nm_aggression_to_othe
  #Plotting previous variable to female visits
  ggplot(aes(total_nm_agg, total_female_visits)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth() + #adds a trendline for the plot
  labs(x = 'Total Nesting Male Aggression Behaviors',
       y = 'Total Female Visits',
       title = 'Total Visits as Function of Total Nesting Male Aggression') + #Makes labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

fvisits.nmagg
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



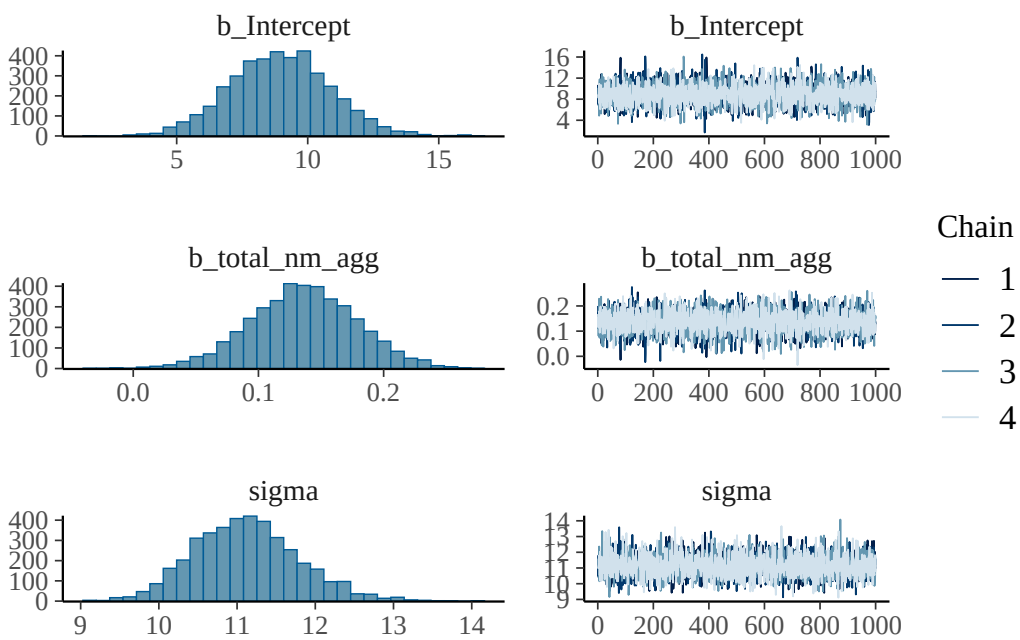
Female visits as a function of Nesting Male aggression model

```
fvisits.nmagg <- behavior_sheet_cln |>
  #creates a variable that is a total of all NM agression
  mutate('total_nm_agg' = nm_aggression_to_sat + nm_aggression_to_snk + nm_aggression_to_othe

fvisits.nmagg.mdl <-
  brm(data = fvisits.nmagg, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ total_nm_agg, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/fvisits.nmagg.mdl') #saves it as an output

summary(fvisits.nmagg.mdl)
```

```
 Family: gaussian
  Links: mu = identity
Formula: total_female_visits ~ total_nm_agg
   Data: fvisits.nmagg (Number of observations: 139)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000
```

```
Regression Coefficients:
             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept        8.95      1.92     5.25    12.75 1.00     3962     2942
total_nm_agg     0.14      0.04     0.05     0.22 1.00     3975     3052

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma    11.12      0.67     9.91    12.52 1.00     4019     3300

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(fvisits.nmagg.mdl)
```

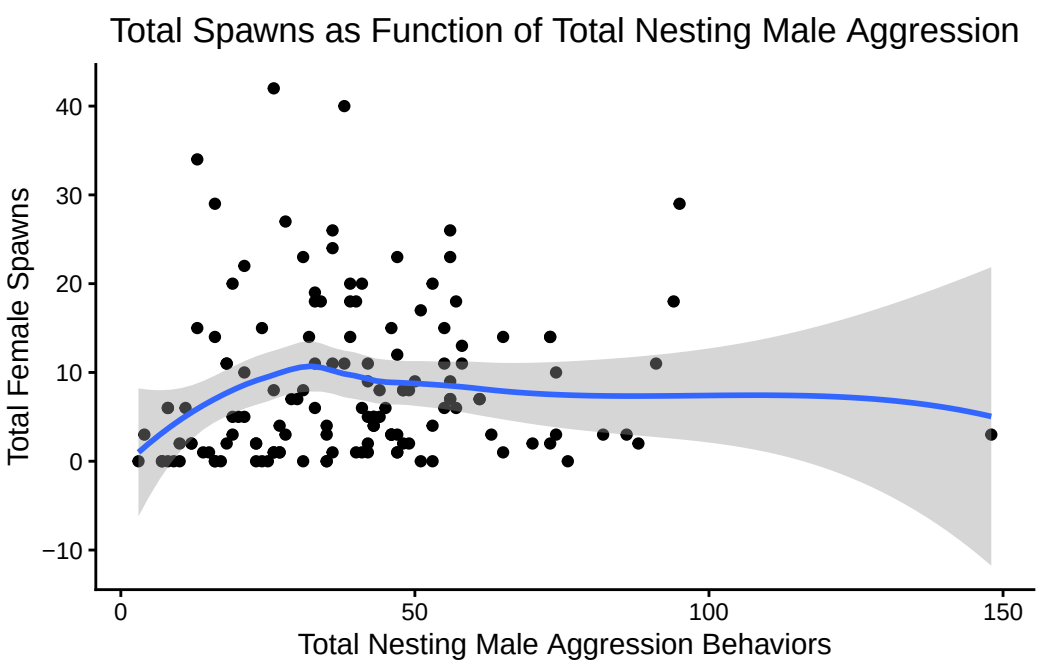


Given our model, we can suggest that total nesting male aggression does have a different from zero effect on total female visits to the nest. Biologically, this does make sense. A more active Nesting Male is a sign of an active nest for a female, additionally, more nesting male movement increases the chances that a female spots a nest.

Female spawns as a function of Nesting Male aggression

```r
f.spawns.nmagg <- behavior_sheet_cln |>
  #creates a variable that is a total of all NM agression
  mutate('total_nm_agg' = nm_aggression_to_sat + nm_aggression_to_snk + nm_aggression_to_othe
  #Plotting previous variable to female visits
  ggplot(aes(total_nm_agg, total_spawns)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth() + #adds a trendline for the plot
  labs(x = 'Total Nesting Male Aggression Behaviors',
       y = 'Total Female Spawns',
       title = 'Total Spawns as Function of Total Nesting Male Aggression') + #Makes labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

f.spawns.nmagg
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```

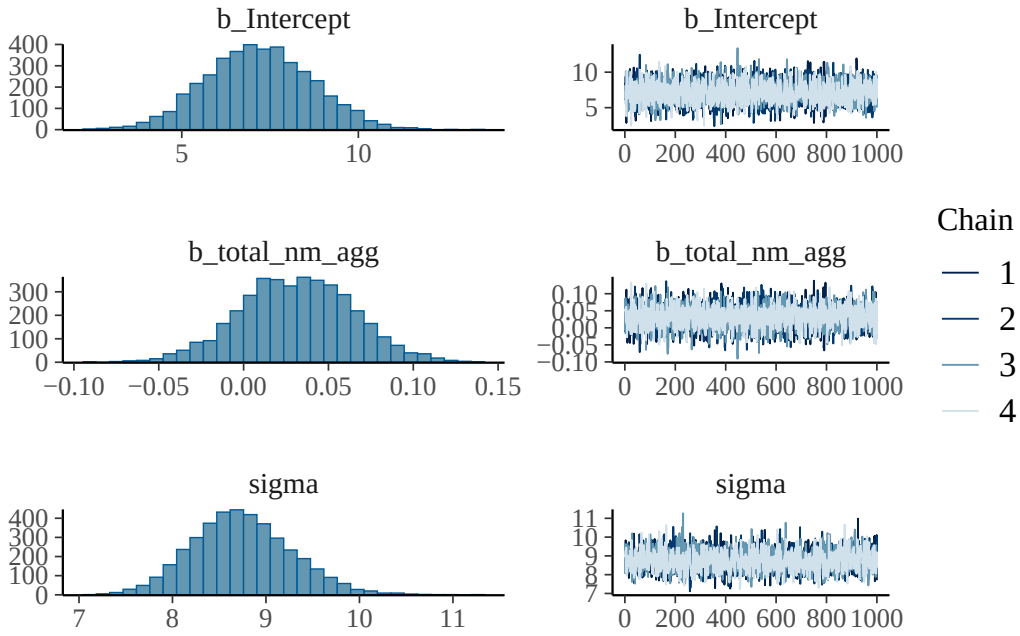


Female spawns as a function of Nesting Male aggression model

```
f.spawns.nmagg <- behavior_sheet_cln |>
  #creates a variable that is a total of all NM agression
  mutate('total_nm_agg' = nm_aggression_to_sat + nm_aggression_to_snk + nm_aggression_to_othe

f.spawns.nmagg.mdl <-
  brm(data = f.visits.nmagg, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_spawns ~ total_nm_agg, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/f.visits.nmagg.mdl') #saves it as an output

summary(f.spawns.nmagg.mdl) #Gives a summary of the model
```

```
 Family: gaussian
  Links: mu = identity
Formula: total_spawns ~ total_nm_agg
   Data: f.visits.nmagg (Number of observations: 139)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept        7.19      1.51     4.31    10.12 1.00     3929     2771
total_nm_agg     0.03      0.03    -0.03     0.09 1.00     3744     2769

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     8.74      0.53     7.77     9.84 1.00     3367     2849

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(f.spawns.nmagg.mdl) #Plots the chains
```



Given our model, we can suggest that Nesting male aggression does not have a different from zero effect on total spawns at a nest. This can be explained biologically. A more active Nesting Male has to spend greater time executing the aggressive behaviors, which means he has less time to spawn at the nest, which could possibly explain the models lower 95% credibility interval of -0.03 (i.e. a nesting male spent so much time with aggressive behaviors that a female chose to leave before spawning).

NOTE:

These models are something to think on more. It seems productive for a female to go to a nest with an active aggressive nesting male, since it indicates mating success, however, there is a point at which he is spending too much time being aggressive that females leave without spawning. Is this because the females no longer want to spawn with him for being too aggressive, or is it because they had to wait to long? Is a female be able to tell the "amount" of active a nesting male is before she visits the nest?

Satellite Male aggression effect on Female Visits and Spawns

This adds up all the columns related to Satellite male aggression (Except aggression towards/away from nest, this will be graphed later) and looks for any correlation with female visits and spawns

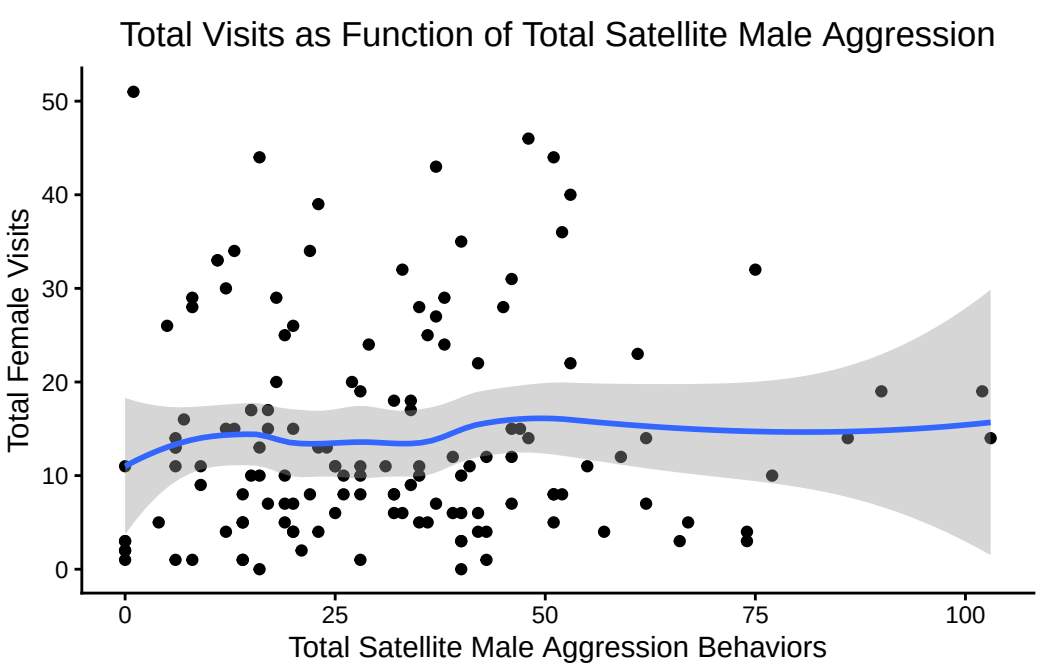
Female visits as a function of Satellite Male aggression

```
f.visits.satagg <- behavior_sheet_cln |>
  ggplot(aes(sat_aggression_to_snk, total_female_visits)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth() + #adds a trendline for the plot
  labs(x = 'Total Satellite Male Aggression Behaviors',
       y = 'Total Female Visits',
       title = 'Total Visits as Function of Total Satellite Male Aggression') + #Makes labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

f.visits.satagg
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```

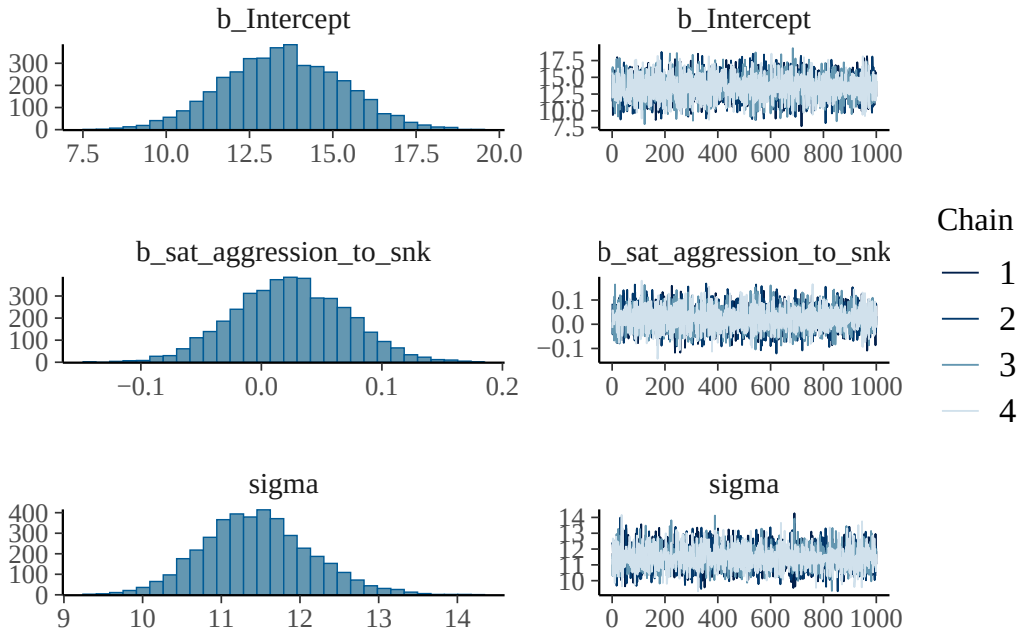


Female Visits as a function of Satellite Male aggression

```
f.visits.satagg.mdl <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ sat_aggression_to_snk, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/f.visits.satagg.mdl') #saves it as an output

summary(f.visits.satagg.mdl)
```

```
 Family: gaussian
  Links: mu = identity
Formula: total_female_visits ~ sat_aggression_to_snk
   Data: behavior_sheet_cln (Number of observations: 140)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
Intercept                13.51      1.77    10.06    16.96 1.00     4523
sat_aggression_to_snk     0.02      0.05    -0.07     0.12 1.00     4369
                      Tail_ESS
Intercept                 3180
sat_aggression_to_snk     2841

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma    11.45      0.69    10.17    12.89 1.00     4486     3016

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(f.visits.satagg.mdl)
```



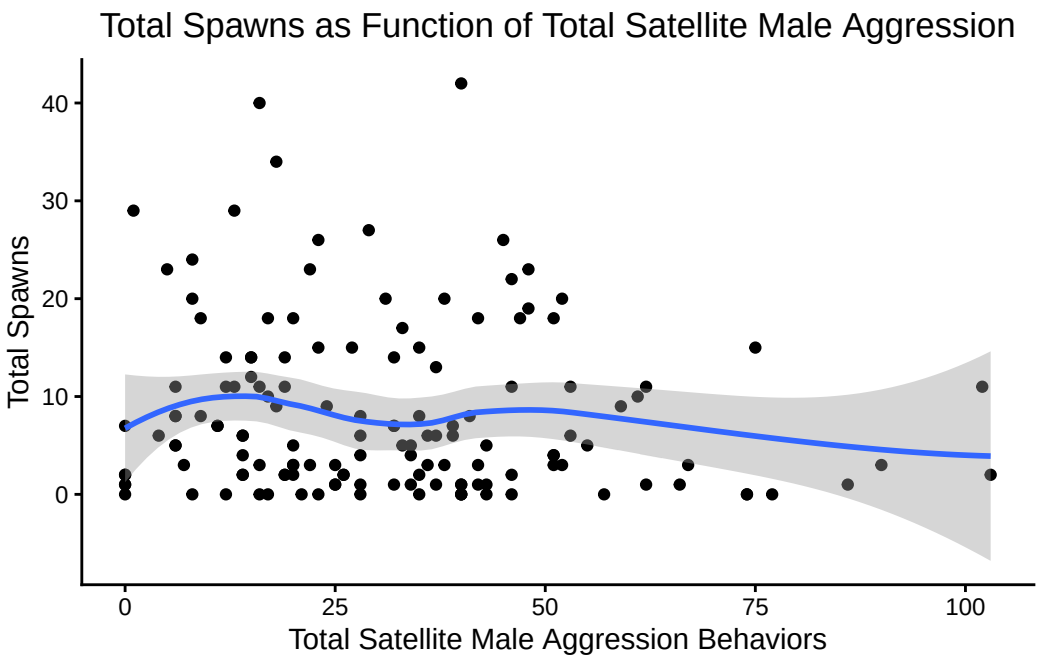
Given our model, we can suggest that sat aggression behaviors effect on female visits to a nest are not different from zero. This could make sense. If a SAT is more focused on being aggressive to sneakers, then he is spending less time focusing on females, thus his behavior has no real effect on female visits. Like our model gives us (albeit not significant in any way) we would expect a small positive effect, since more sat aggressive behavior means more sneakers which indicates to females nest activity.

Female spawns as a function of Satellite Male aggression

```
f.spawns.satagg <- behavior_sheet_cln |>
  ggplot(aes(sat_aggression_to_snk, total_spawns)) + #gives ggplot dimensions
  geom_point() + #makes the plot a scatterplot
  geom_smooth() + #adds a trendline for the plot
  labs(x = 'Total Satellite Male Aggression Behaviors',
       y = 'Total Spawns',
       title = 'Total Spawns as Function of Total Satellite Male Aggression') + #Makes labels
  theme_classic() + #Sets the plot visuals
  theme(plot.title = element_text(hjust = 0.5)) #Moves title to the middle

f.spawns.satagg
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```



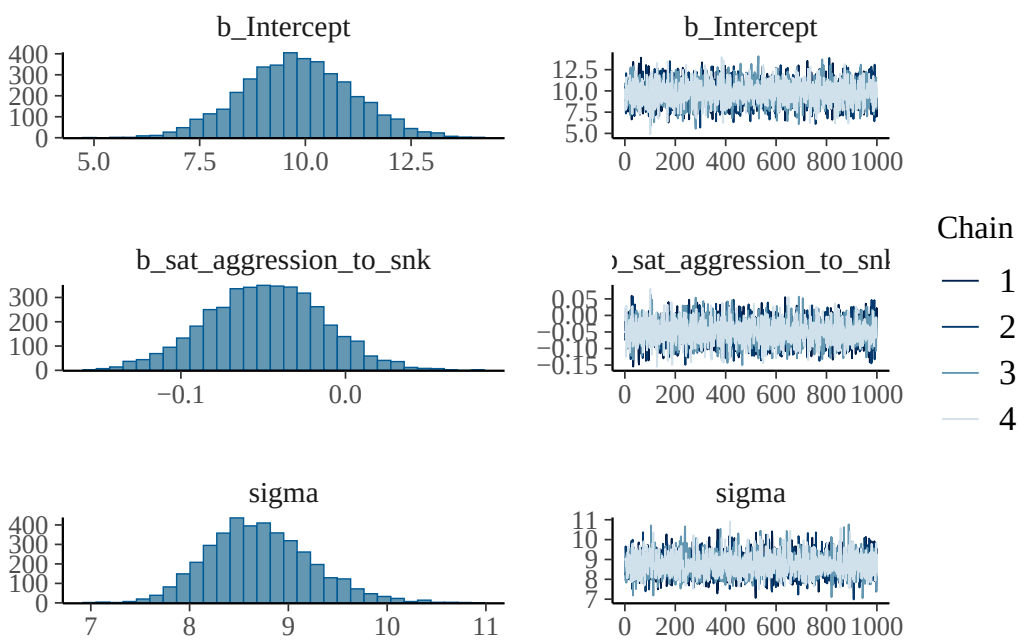
Female spawns as a function of Satellite male aggression model

```
f.spawns.satagg.mdl <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_spawns ~ sat_aggression_to_snk, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/f.spawns.satagg.mdl') #saves it as an output

summary(f.spawns.satagg.mdl)
```

```
 Family: gaussian
  Links: mu = identity
Formula: total_spawns ~ sat_aggression_to_snk
```

```
   Data: behavior_sheet_cln (Number of observations: 140)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
Intercept                 9.81      1.31     7.27    12.36 1.00     4141
sat_aggression_to_snk    -0.05      0.04    -0.12     0.02 1.00     4110
                      Tail_ESS
Intercept                 2880
sat_aggression_to_snk     2928

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     8.72      0.53     7.78     9.86 1.00     4222     2761

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(f.spawns.satagg.mdl)
```



Like previous, given our model Satellite male aggressive behaviors effect of total spawns at a

nest are not different from zero. However, the models estimate of -0.05 is interesting. Say the model was different from zero, then there would be a negative effect of satellite male aggression on total spawns. Why would this be the case for Satellite males and not Nesting males? Is it because if a satellite is being more aggressive that means the nest is too active and the females don't want to spawn with so many sneakers around? Something to think about.

Satellite and Nesting Male aggression to Females, away and towards the nest

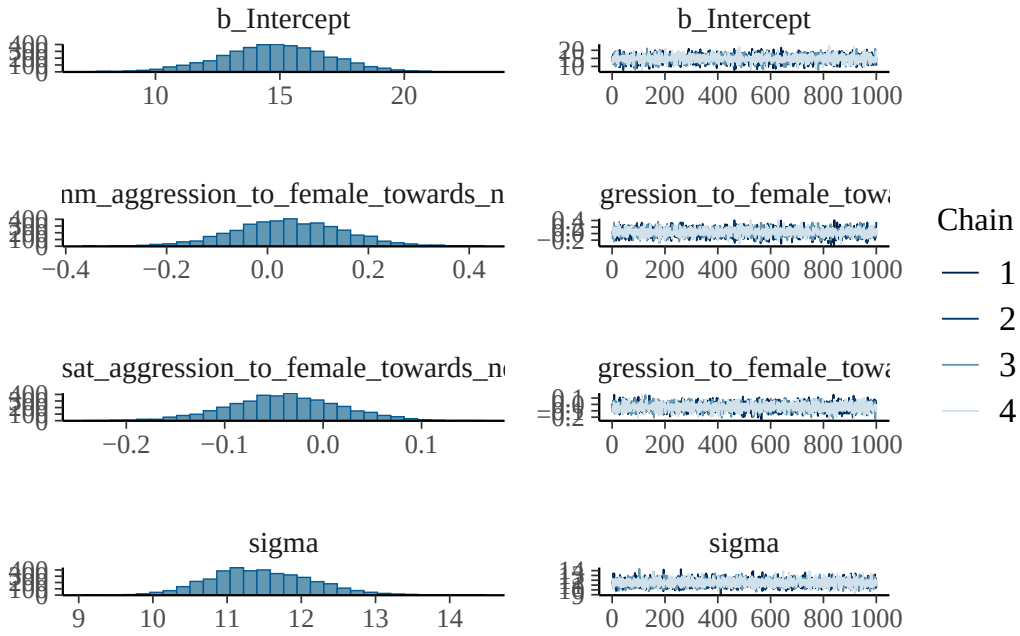
We would expect any aggression to have negative impacts on female spawns, possibly also visits. (Note, if different than zero effects are found, it might be worth parsing which male has the greater effect).

Female visits as a function of Nesting and Satellite aggression towards the nest

```
f.v.nestsat.agg2nest <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ nm_aggression_to_female_towards_nest + sat_aggression_to_female_t
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/f.v.nestsat.agg2nest') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```
summary(f.v.nestsat.agg2nest)
plot(f.v.nestsat.agg2nest)
```

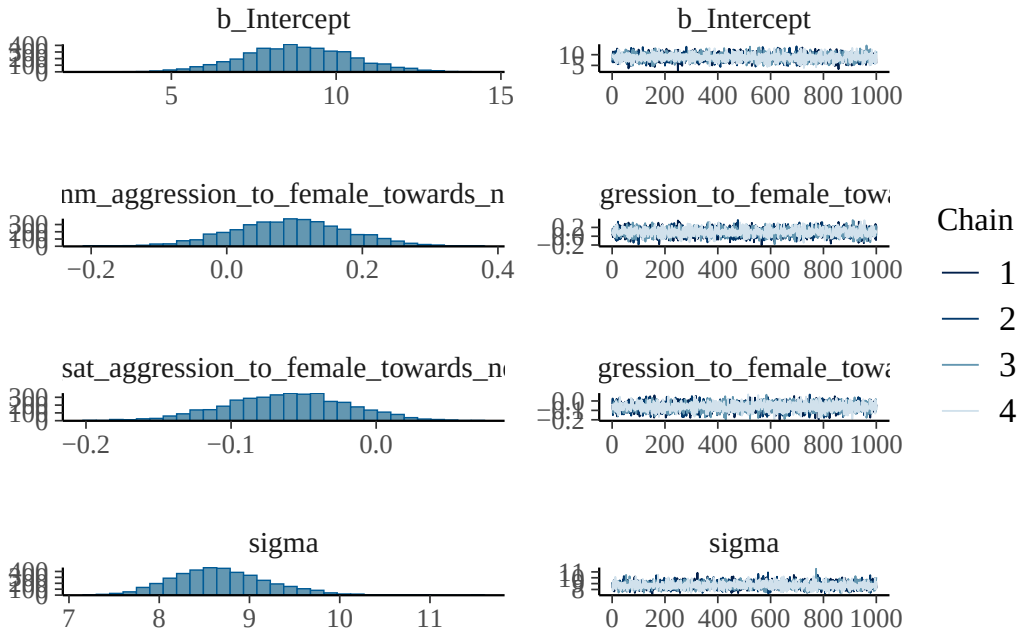


Female spawns as a function of Nesting and Satellite aggression towards the nest

```
f.s.nestsat.agg2nest <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_spawns ~ nm_aggression_to_female_towards_nest + sat_aggression_to_female_towards_
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/f.s.nestsat.agg2nest') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```
summary(f.s.nestsat.agg2nest)
plot(f.s.nestsat.agg2nest)
```

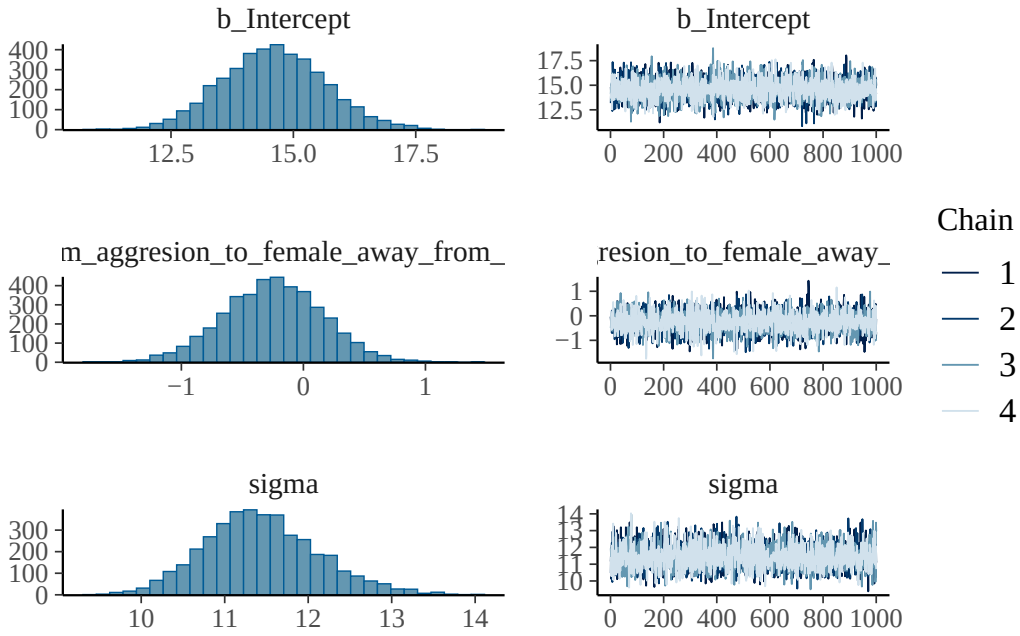


Female visits as a function of Nesting and Satellite aggression away from the nest model (there is no Satellite aggression away from the nest variable)

```
f.v.nestsat.aggawaynest <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ nm_aggresion_to_female_away_from_nest, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/f.v.nestsat.aggawaynest') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```
summary(f.v.nestsat.aggawaynest)
plot(f.v.nestsat.aggawaynest)
```

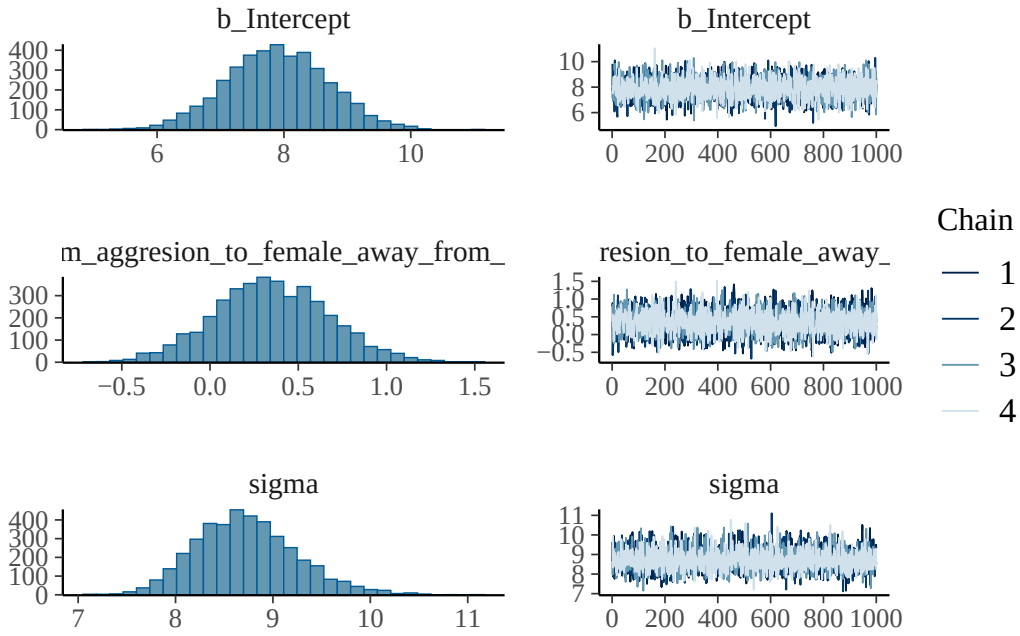


Female spawns as a function of Nesting and Satellite aggression away from the nest model (there is no Satellite aggression away from the nest variable)

```
f.s.nestsat.aggawaynest <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_spawns ~ nm_aggresion_to_female_away_from_nest, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/f.s.nestsat.aggawaynest') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```
summary(f.s.nestsat.aggawaynest)
plot(f.s.nestsat.aggawaynest)
```



Sneaker Presence and Female Visits and Spawns

From previous research it has been found that females use the presence of sneakers as a proxy for Nesting Male success, this graph will observe this.

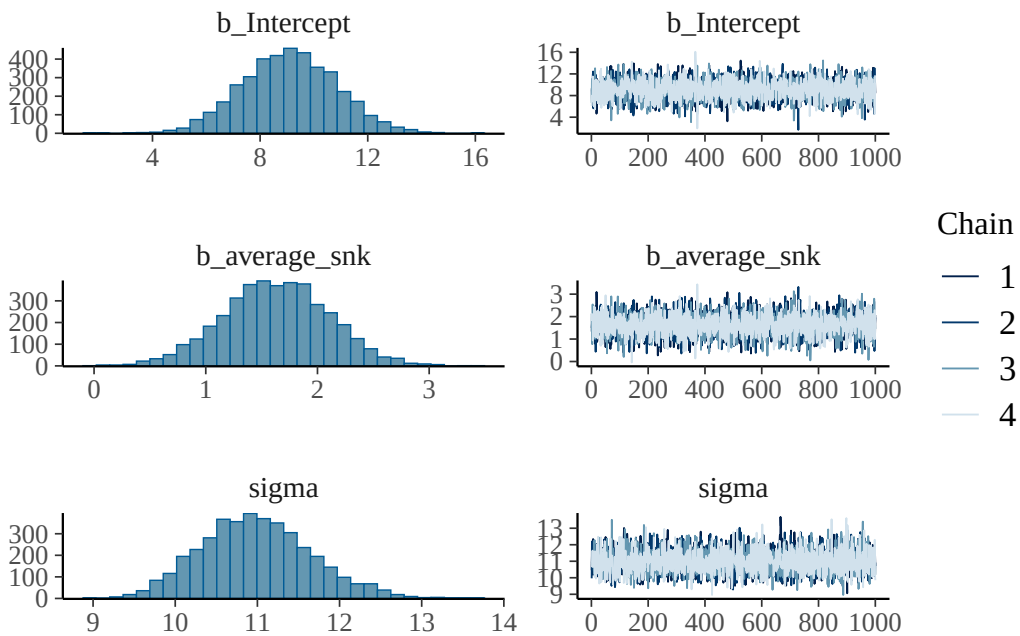
Female visits as a function of number of sneakers at a nest

```
f.v.snkr.at.nest <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ average_snk, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/f.v.snkr.at.nest') #saves it as an output
```

```
Compiling Stan program...
```

```
Start sampling
```

```r
summary(f.v.snkr.at.nest)
plot(f.v.snkr.at.nest)
```

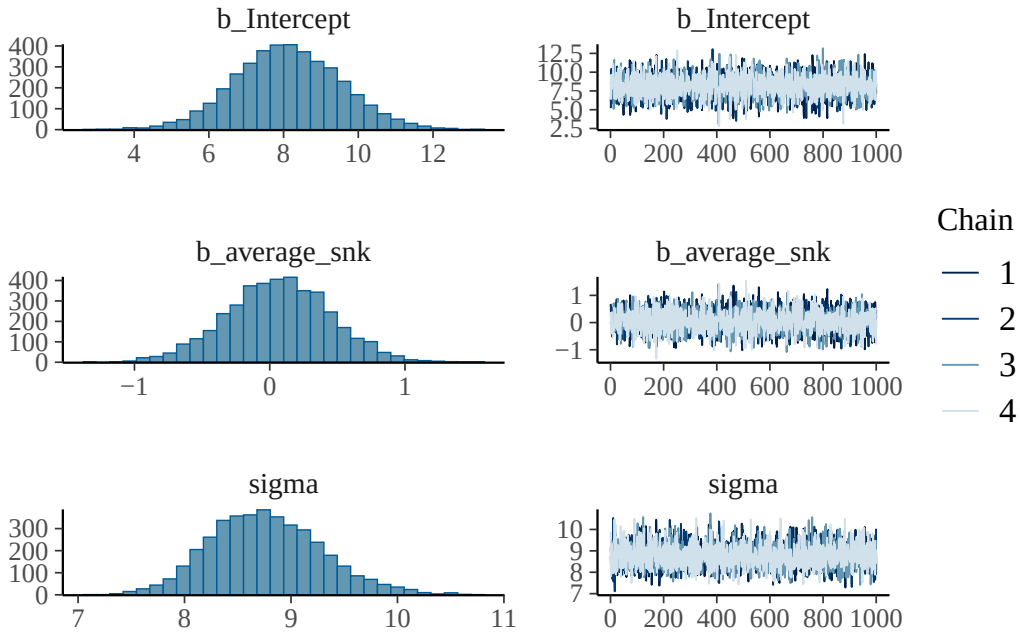


Female spawns as a function of number of sneakers at a nest

```r
f.s.snkr.at.nest <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_spawns ~ average_snk, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/f.s.snkr.at.nest') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```r
summary(f.s.snkr.at.nest)
plot(f.s.snkr.at.nest)
```



Positive interactions between Satellite and Nesting Male and Female Visits and Spawns

This will add up the positive interactions occurring between the Satellite and Nesting Male (female passing)

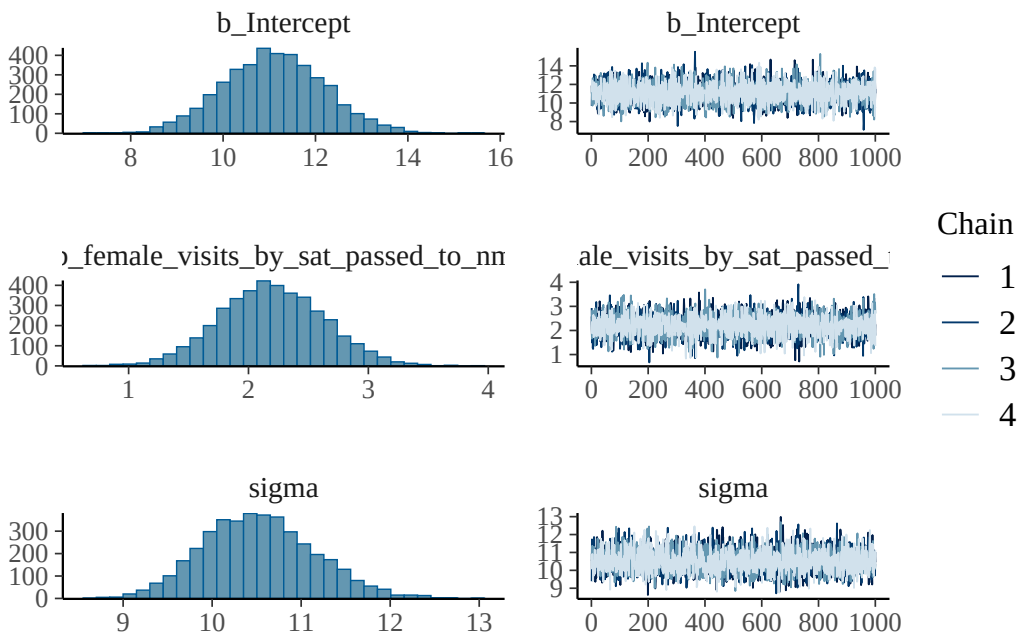
Female visits as a function of Positive interactions between Nesting and Satellite Males

```
fv.pos.nmsat <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ female_visits_by_sat_passed_to_nm , #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/fv.pos.nmsat') #saves it as an output
```

```
Compiling Stan program...
```

```
Start sampling
```

```r
summary(fv.pos.nmsat)
plot(fv.pos.nmsat)
```



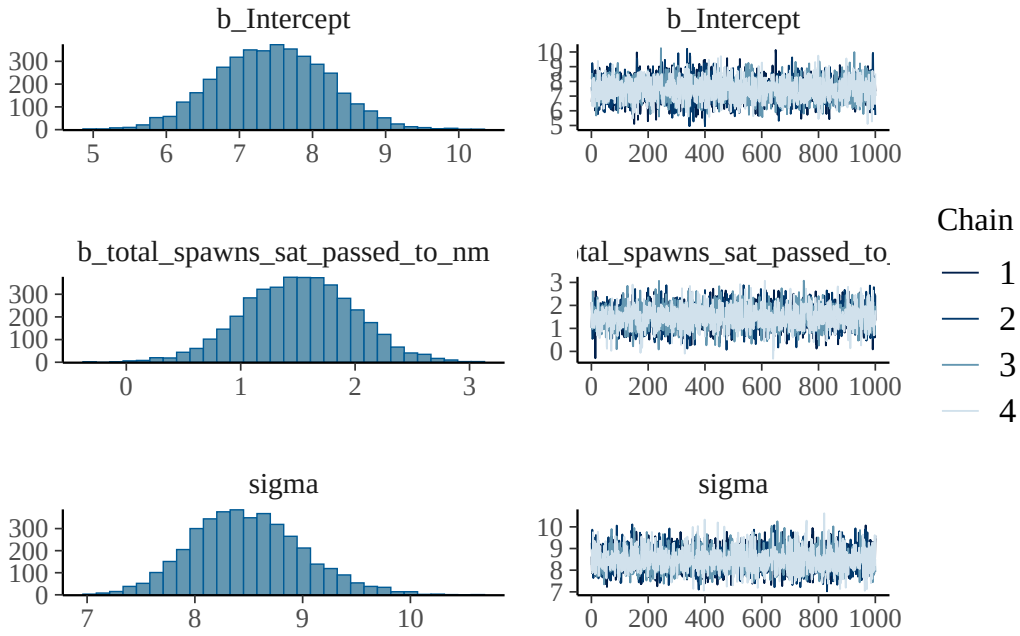
Female spawns as a function of Positive interactions between Nesting and Satellite Males

```r
fs.pos.nmsat <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_spawns ~ total_spawns_sat_passed_to_nm, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/fs.pos.nmsat') #saves it as an output
```

```
Compiling Stan program...
```

```
Start sampling
```

```r
summary(fs.pos.nmsat)
plot(fs.pos.nmsat)
```



Negative interactions between Satellite and Nesting Male and Female Visits and Spawn

This does the same above, except with the negative interactions between Satellite and Nesting Male (SAT Submission to NM, NM aggression to SAT).

Female visits as a function of Negative interactions between Nesting and Satellite Males

Female spawns as a function of Negative interactions between Nesting and Satellite Males

Satellite and Nesting male interactions against sneakers effect on Female Visits and Spawns

Female prefer to not spawn with sneakers, thus higher Satellite and Nesting male aggression towards sneakers should increase the number of spawns. However, too many sneakers to chase means not enough time to spawn. This graph will let us see the relationship.

Female visits as a function of Nesting and Satellite male interactions against sneakers

Female spawns as a function of Nesting and Satellite male interactions against sneakers

Nesting male courtship and Female Visits and Spawns

Female visits as a function of Nesting male courtship

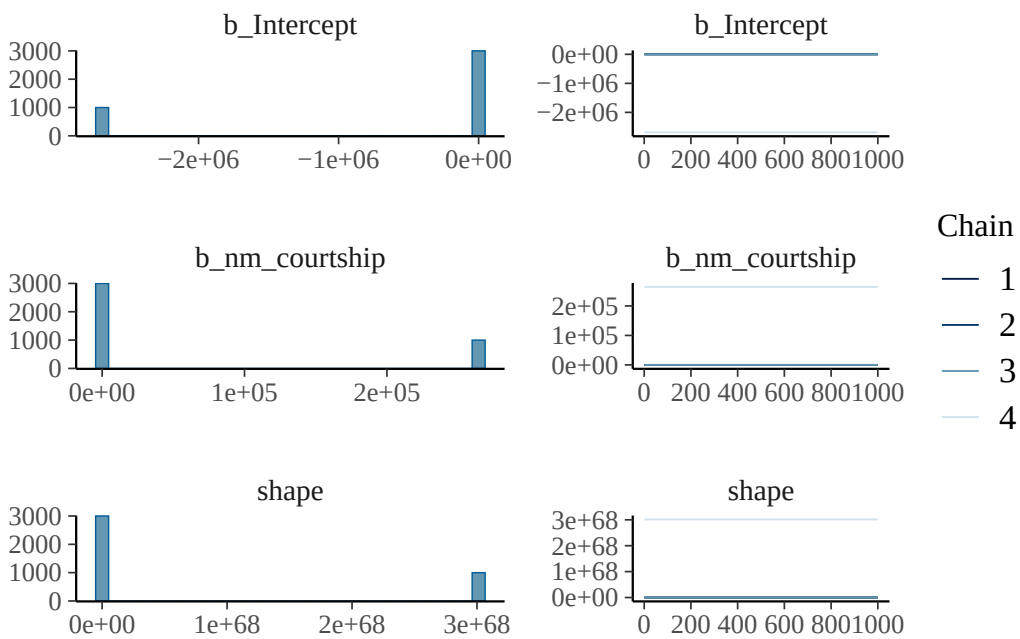
```
fv.nm.court1 <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = negbinomial(), #Tell model what type of distribution
      total_female_visits ~ nm_courtship, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/fv.nm.court2') #saves it as an output
```

```
Compiling Stan program...

Start sampling

Warning: There were 65 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: There were 935 transitions after warmup that exceeded the maximum treedepth. Increas
https://mc-stan.org/misc/warnings.html#maximum-treedepth-exceeded

Warning: Examine the pairs() plot to diagnose sampling problems

Warning: The largest R-hat is 1.53, indicating chains have not mixed.
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#r-hat

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and median
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess

Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and ta
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess
```

```r
summary(fv.nm.court1)
```

```
Warning: Parts of the model have not converged (some Rhats are > 1.05). Be
careful when analysing the results! We recommend running more iterations and/or
setting stronger priors.

Warning: There were 65 divergent transitions after warmup. Increasing
adapt_delta above 0.8 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

```r
plot(fv.nm.court1)
```

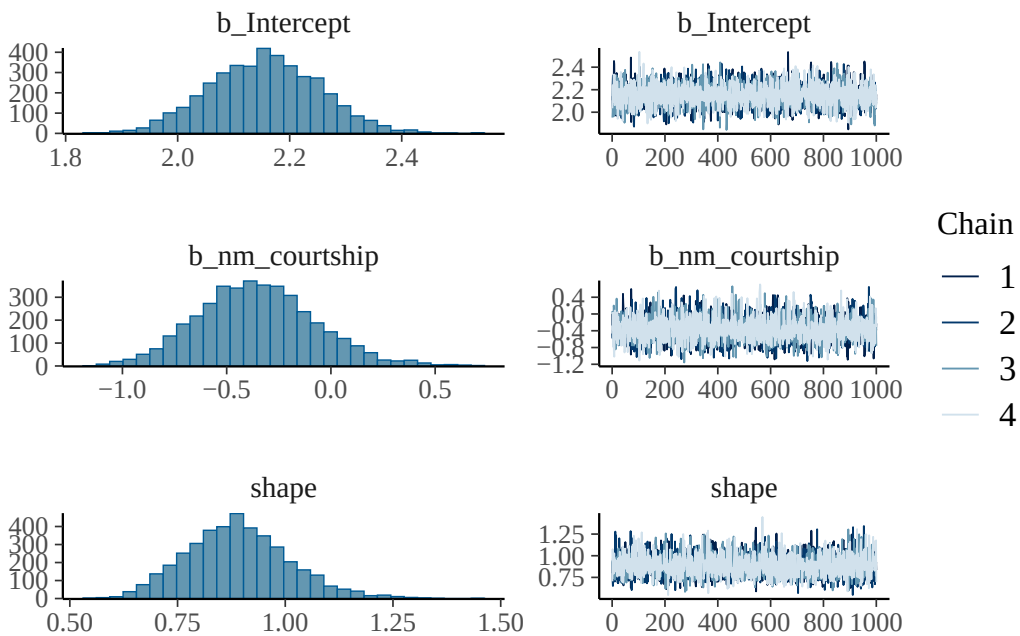


Female spawns as a function of Nesting male courtship

```r
fs.nm.court1 <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = negbinomial(), #Tell model what type of distribution
      total_spawns ~ nm_courtship, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/fs.nm.court1') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```
summary(fs.nm.court1)
plot(fs.nm.court1)
```



Does it matter who brings the female?

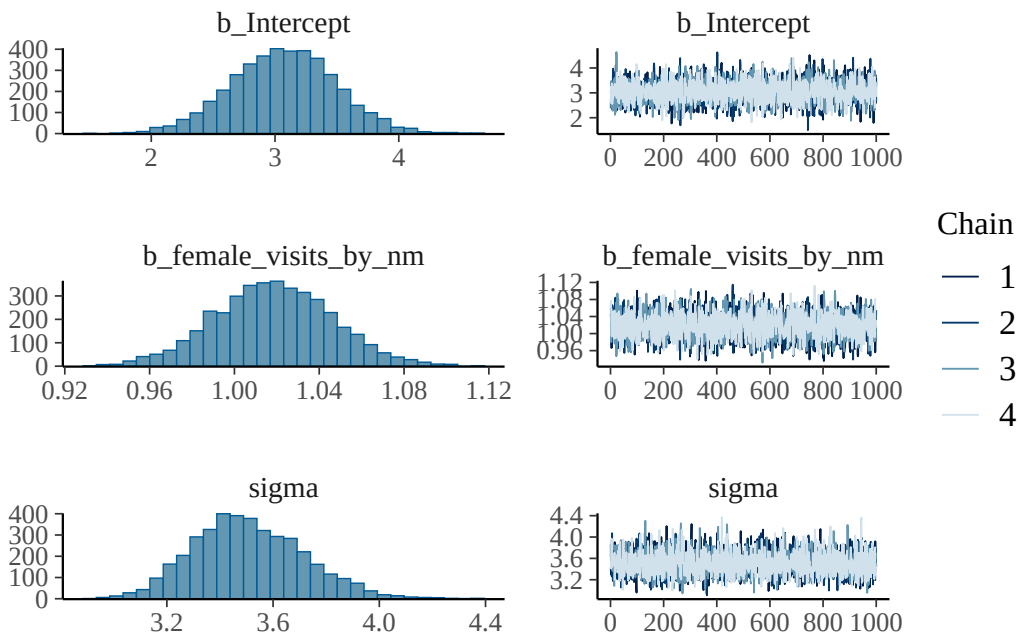
These graphs will look at the relationship between which male type brings the female and female spawning.

Female visits as a function of being brought by Nesting male

```
fv.brought.by.nm <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ female_visits_by_nm, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/fv.brought.by.nm') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```r
summary(fv.brought.by.nm)
plot(fv.brought.by.nm)
```

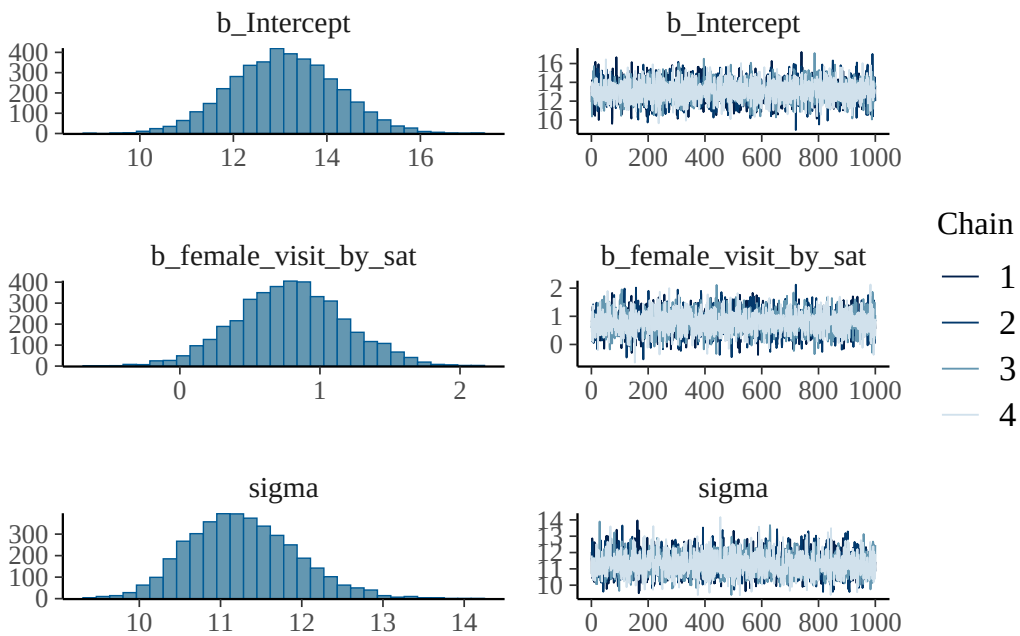


Female visits as a function of being brought by Satellite male

```r
fv.brought.by.sat <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ female_visit_by_sat, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/fv.brought.by.sat') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```r
summary(fv.brought.by.sat)
plot(fv.brought.by.sat)
```

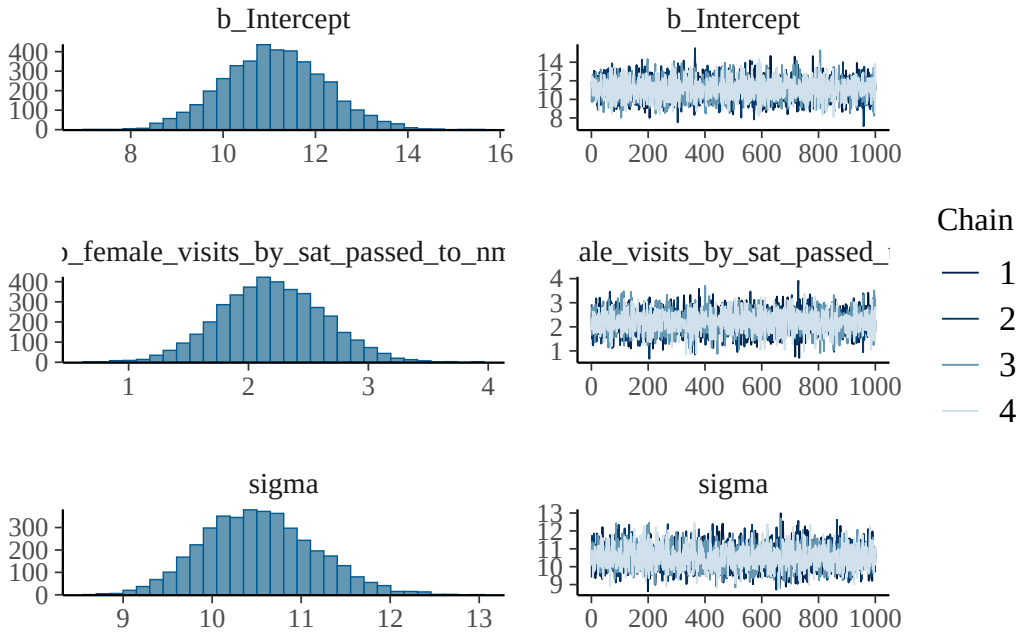


Female visits as a function of being brought by Satellite male and passed to Nesting male

```r
fv.brought.by.sat.passed2nm <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ female_visits_by_sat_passed_to_nm, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/fv.brought.by.sat.passed2nm') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```r
summary(fv.brought.by.sat.passed2nm)
plot(fv.brought.by.sat.passed2nm)
```

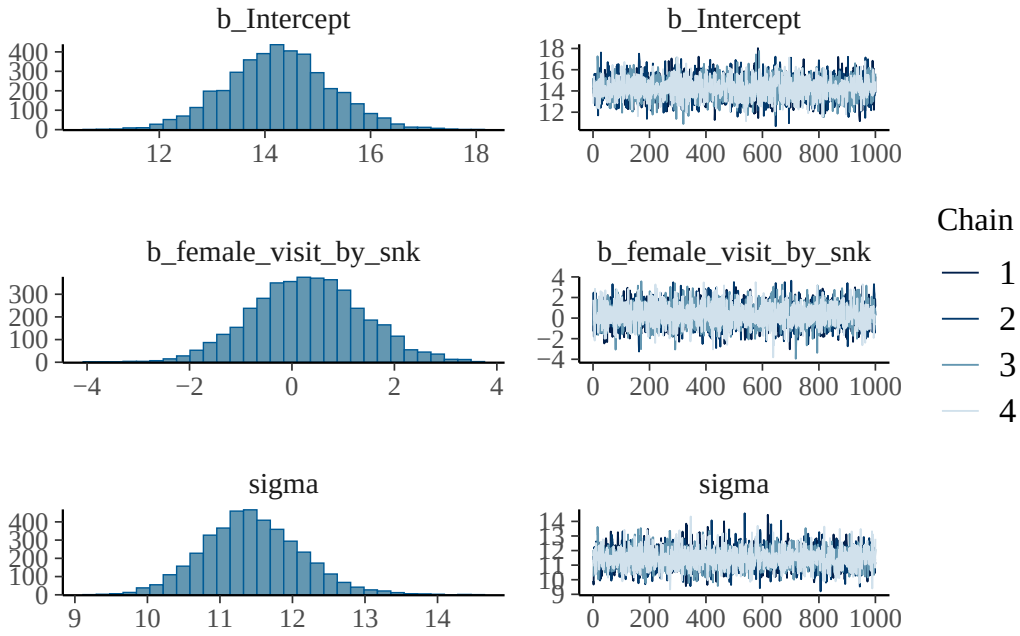


Female visits as a function of being brought by a sneaker male

```
fv.brought.by.snkr <-
  brm(data = behavior_sheet_cln, #Give the model the tibble
      family = gaussian, #Tell model what type of distribution
      total_female_visits ~ female_visit_by_snk, #Specifying the model
      iter = 2000, warmup = 1000, chains = 4, cores = 4, #Markov chain parameters
      seed = 4, #Determines which random numbers
      file = 'Exploratory model outputs/fv.brought.by.snkr') #saves it as an output
```

```
Compiling Stan program...

Start sampling
```

```
summary(fv.brought.by.snkr)
plot(fv.brought.by.snkr)
```



Building/looking at our 'final' model.

Overarching question: How does the social environment influence female mate choice?

target outcome: How many females decided to spawn

Thus total_female_spawning is the our response variable, because it tells us how many females made the choice to spawn.

Based on our current understanding (and data), what has *direct* effects on this response?

1. Average number of sneaker
2. Social interactions at the nest

    1. NM to SAT
    2. NM to SNK
    3. SAT to SNK
    4. NM to F
    5. SAT to F

How are these factors mentioned in our data.

1. average_snk
2. Social interactions at the nest

1. female_visits_by_sat_passed_to_nm, female_spawning_sat_passed_to_nm, female_spawning_sat, nm_aggression_to_sat, sat_submission_to_nm
2. nm_aggression_to_snk, snk_submission_to_nm
3. sat_aggression_to_snk, snk_submission_to_sat
4. female_visits_by_nm, female_spawning_with_nm_brought_by_nm, nm_aggression_to_female_ nm_aggression_to_female_away_from_nest
5. female_visit_by_sat, female_spawning_sat_passed_to_nm_but_sat_only_spawn, females_spawing_sat_brought_sat_only_spawn, sat_aggression_to_female_towards_nest

Note: It is likely that nm_aggression_to_snk, snk_submission_to_nm, sat_aggression_to_snk, snk_submission_to_sat would be statistically the same as average_snk as each one of those are a related to how many sneakers are present.

Linear regression models

total_female_spawning ~ average_snk

total_female_spawning ~ female_visits_by_sat_passed_to_nm

total_female_spawning ~ female_spawning_sat_passed_to_nm

total_female_spawning ~ female_spawning_sat

total_female_spawning ~ nm_aggression_to_sat

total_female_spawning ~ sat_submission_to_nm

total_female_spawning ~ nm_aggression_to_snk

total_female_spawning ~ snk_submission_to_nm

total_female_spawning ~ sat_aggression_to_snk

total_female_spawning ~ snk_submission_to_sat

total_female_spawning ~ female_visits_by_nm

total_female_spawning ~ female_spawning_with_nm_brought_by_nm

total_female_spawning ~ nm_aggression_to_female_towards_nest

total_female_spawning ~ nm_aggression_to_female_away_from_nest

total_female_spawning ~ female_visit_by_sat

total_female_spawning ~ female_spawning_sat_passed_to_nm_but_sat_only_spawn

total_female_spawning ~ females_spawing_sat_brought_sat_only_spawn

total_female_spawning ~ sat_aggression_to_female_towards_nest

However, these models are unrealistic, since it is highly unlikely that each one of these factors influence female spawning decision alone. Thus it is important for us to

Multiple Regression models

ggcore? to look at how variables covary?

For sneaker related models

total_female_spawning ~ nm_aggression_to_snk + snk_submission_to_nm + sat_aggression_to_snk + snk_submission_to_sat

Note: The associated aggression and submission variables (nm and sat respectively) are likely to be redundant since submission is a response to aggression, thus it is likely that both might not be needed. We will run both models to check.

For Nesting Male/SAT related models

total_female_spawning ~ female_visits_by_sat_passed_to_nm + female_spawning_sat_passed_to_nm + female_spawning_sat +nm_aggression_to_sat + total_female_spawning ~ sat_submission_to_nm

For NM to F

total_female_spawning ~ female_visits_by_nm + female_spawning_with_nm_brought_by_nm + nm_aggression_to_female_towards_nest + nm_aggression_to_female_away_from_nest

For SAT to F

total_female_spawning ~ female_visit_by_sat + female_spawning_sat_passed_to_nm_but_sat_only_spaw + females_spawing_sat_brought_sat_only_spawn + sat_aggression_to_female_towards_nest

After meeting with Malin

The multiple regression model(s) should be a binomial with the N being the number of female visits, and p being the probability of spawning given a visit. We should normalize the pathways (variables) before putting them in the model, meaning take each variable as a fraction of visits for each pathways, i.e. fraction of females that spawned which were brought by the SAT, fraction of females that spawned which were brought by the SAT and passed to the NM.

To-Dos:

Look at the data distribution for the effect of sneakers on female visits and female spawning. Biologically, quadratic makes sense (low sneakers = low nest success = female doesn't visit/spawn, medium sneakers = moderate nest success = female likely visits/spawns, high sneakers = high nest success = females don't spawn due to high sneaker numbers.

Normalize the variables that we think are important, "female_visits_by_nm, female_visit_by_sat, female_visits_by_sat_passed_to_nm" by the total number of visits

```
#bf(spawns|trails(visits) ~ 1 + )
```